AI- DRIVEN SMART CINNAMON GROWTH MONITORING & AUTOMATED HARVESTING SYSTEM

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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ABSTRACT

One of the economic agricultural crops is cinnamon. In order to grow it in a healthy way and cultivate productively, there is a need for certain environmental and soil conditions. Traditional agriculture of the investigated crop usually implies observation and measurements done manually, which might not always be sufficient due to its limitations of providing continuous and up-to-date information. Internet of Things technology can be applied in order to ensure continuous monitoring

and provision of reliable information about environmental changes. This work introduces IoT-based monitoring and sensing system as one of the components of the proposed AI-Driven Multi-Environment Cinnamon Growth Monitoring and Automated Harvesting System Using Robotic Manipulation and Multispectral Sensing.

As one of the main goals of the proposed system, the development of the IoT component aimed at gathering environmental and soil-related data in real time and in multiple cinnamon cultivation environments. ESP32 microcontrollers have been used as the processing and communication devices in the proposed distributed system of sensors. Sensor nodes contain environmental sensors to measure such parameters as temperature, humidity, soil moisture, light/UV conditions, etc. The measurements are processed by the ESP32 microcontroller and transferred wirelessly to the central data collector through the communication interface. ESP-NOW communication method is used to ensure efficient and lightweight communication between sensor nodes and the central data collector device.

The information about measurements is received and organized in the central data collector, making the collected data available for real-time monitoring and further analysis. IoT component can be used for the whole harvesting system, providing the necessary sensor data related to the harvesting environment. The collected information might become useful for further AI and machine learning modules in such tasks as plant growth monitoring and predictions, harvest-related decisions.

The system has been implemented and evaluated for the purposes of testing data acquisition process, wireless communication, data collection and monitoring functions. The obtained results prove the possibility of applying the ESP32-based IoT infrastructure for continuous multi-environment monitoring of cinnamon cultivation.

Keywords: Internet of Things (IoT), ESP32, Cinnamon Cultivation, Environmental Monitoring, Soil Moisture Monitoring, ESP-NOW, Sensor Networks, Real-Time Data Collection, Smart Agriculture.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
AI/ML	Artificial Intelligence and Machine Learning

Abbreviation	Meaning
CPU	Central Processing Unit
ESP32	Espresso Systems 32-bit Microcontroller Series
ESP-NOW	Espresso Wireless Communication Protocol
Firebase	Google Firebase Application Development Platform
ML	Machine Learning
OLED	Organic Light-Emitting Diode
UART	Universal Asynchronous Receiver-Transmitter

1.INTRODUCTION

1.1 Overall Project Context

Cinnamon is a significant economic agricultural product of Sri Lanka, which contributes significantly to the agricultural industry of the country as well as its export market. However, traditional methods of cultivating and harvesting cinnamon rely mostly on the personal experience of the farmers and the observation made manually. Therefore, constant monitoring of the health condition, environment, and other factors might become challenging. Hence, the application of

new modern technologies such as Internet of Things (IoT), Artificial Intelligence (AI), Machine Learning (ML), multispectral sensing, and robotics can help to build a smarter and more efficient system for cultivation and harvesting of cinnamon.

This project, AI-Driven Multi-Environment Cinnamon Growth Monitoring & Automated Harvesting System Using Robotic Manipulation and Multispectral Sensing, aims at developing an integrated system that would monitor cinnamon plant growth under different cultivation environments and support automated harvesting. The system includes several components: IoT environmental monitoring, disease detection, growth monitoring and harvesting readiness prediction, multispectral sensing, and automated cinnamon harvesting machine.

IoT-based part of the system constantly measures important environmental factors such as temperature, humidity, moisture content of the soil, light or UV conditions with the help of ESP32-based sensor nodes. Moreover, there is a possibility to transfer sensor data through wireless communication (ESP-NOW). Measurement of environmental factors can serve as additional data for assessing plant growth and condition.

In addition to this, the system employs AI and ML models for cinnamon disease detection with the help of which it will be possible to analyze images of the plant to detect any diseases or abnormal conditions. Moreover, ML growth monitoring and prediction models will be able to assess how the plant is growing. Besides, the harvesting readiness model will estimate the stage when harvesting should take place. Additionally, multispectral sensing will help to capture data of the plant from various wavelengths, which could be useful for estimating the condition of the plant.

Lastly, the automated cinnamon harvesting machine will incorporate robotics and sensors to harvest cinnamon based on the information collected by the previous system components. Thus, the overall system employs IoT, AI, ML, multispectral sensing, and robotics to develop a data-driven solution for monitoring, identifying diseases, assessing growth, making harvesting decision, and harvesting cinnamon automatically.

1.2 Individual Component Context

Internet of Things (IoT) component is one of the parts of the proposed AI-Driven Multi-Environment Cinnamon Growth Monitoring and Automated Harvesting System. The main aim of this individual component is to provide the environmental monitoring and communication layer of the system by constantly collecting important environmental and soil parameters that might affect the process of growing, cultivating and harvesting the cinnamon plants. Using the IoT component, it is possible to replace periodic human observations with automatic monitoring of the cultivation environment.

As for design choices, ESP32 microcontrollers are used in the IoT system as the main devices that will be responsible for processing and communication of the data. There are multiple sensor nodes in order to allow observing several cultivation environments at once. Every node includes several sensors that measure the important parameters of the environment like temperature, humidity, soil moisture, and light/UV conditions. In other words, each sensor node includes the following: connected sensors, ESP32 module that receives, processes raw measurements and prepares them for the wireless transmission.

In order to implement the communication between the sensor nodes and the main data collector, ESP-NOW wireless communication protocol is used. With this protocol, each sensor node is able to send the collected measurements to the main collector without the need of establishing a standard Wi-Fi network connection between each node and the data collector. Main collector collects the measurements sent by each distributed node, separates and organizes the received data and prepares it for further processing.

In turn, the collected IoT data is very helpful in the analysis of cinnamon plant growth and conditions together with the data collected by other parts of the proposed research system. IoT data can be used together with the plant images, multispectral information and AI/ML algorithms' output.

Moreover, the IoT element plays a significant role in creating the necessary background for intelligent decision-making and automated harvesting within the research system as a whole. Data from the environmental parameters gathered by means of the sensor network may be combined with the information obtained from the disease detection, growth prediction, multispectral measurement, and harvesting readiness prediction elements. Thus, it is possible to assess the plants based on several pieces of information rather than just one criterion. It should be acknowledged that the developed IoT element creates the connection between the physical world of cultivation and the digital world of the proposed system.

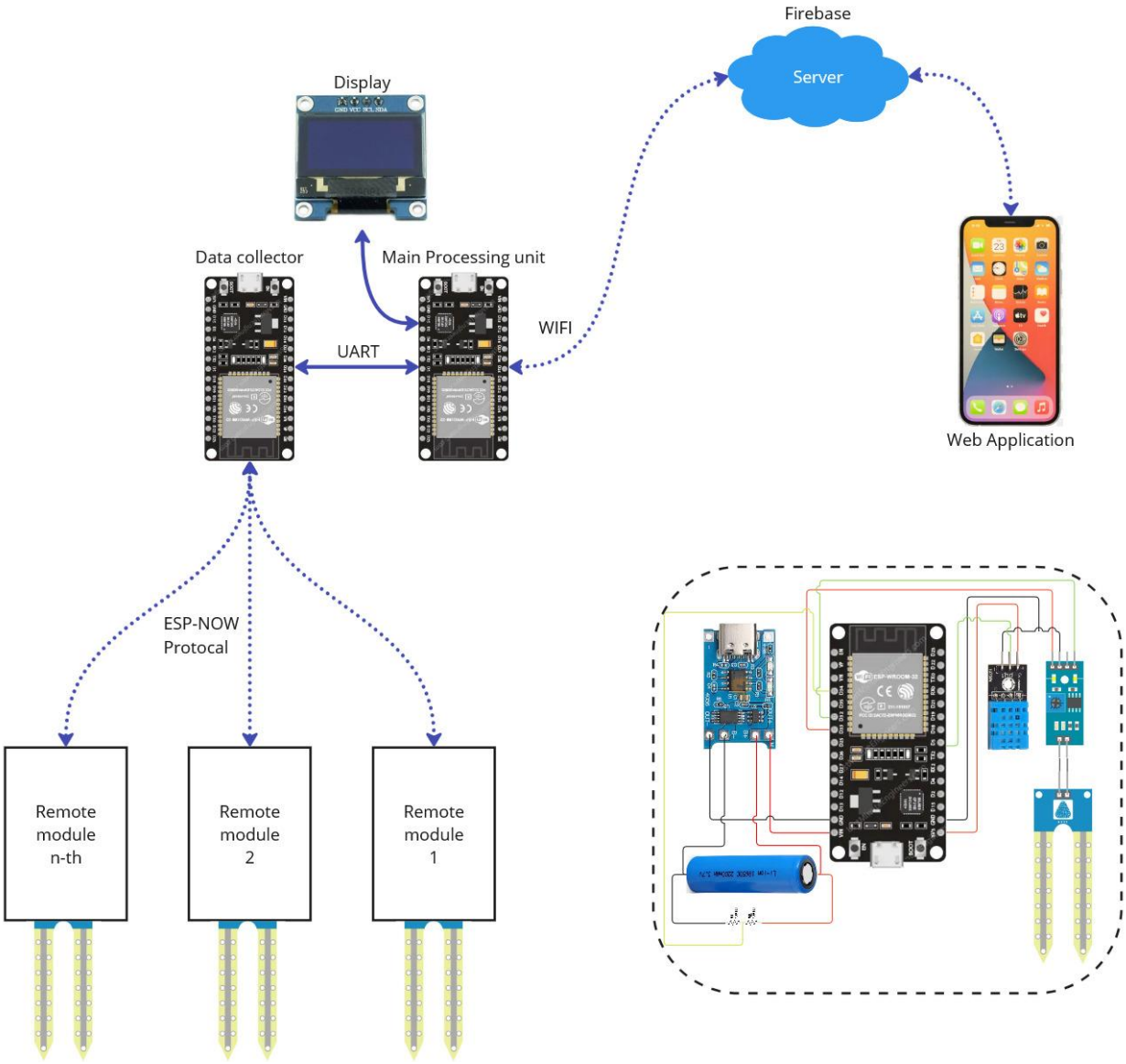


Figure 1Overall System Diagram

1.3 Problem Statement

For proper cinnamon cultivation, the cultivation environment needs to meet specific requirements related to the quality of the surrounding soil and conditions in order to ensure healthy growth and obtain high-quality crops. At the same time, the current practice of monitoring cultivation environment conditions includes only manual observations and measurements conducted periodically by farmers. Temperature, humidity, soil moisture level, and light conditions can vary during the day and, therefore, it is impossible to detect these changes with periodic measurements. As a consequence, farmers will lack continuous information regarding the exact conditions around cinnamon plants in cultivation.

Moreover, there is a problem associated with the impossibility of monitoring several cultivation environments' conditions at once due to the fact that each location will have different temperature, humidity, soil moisture level, and light conditions affecting the growth of cinnamon plants. The manual collection of measurements from several locations requires time and energy investments and results in inconsistent collection interval. In addition, there is no efficient mechanism for transferring collected measurements to the central system for further analysis. Therefore, it becomes difficult to combine collected environmental data with the other types of data, for instance, images, multispectral measurements, disease detection results, and growth predictions.

Finally, the lack of an automated reliable communication mechanism complicates the development of the real-time monitoring system since collected sensor data must be transferred efficiently to the central data collector without increasing communication complexity and power consumption. Thus, a wireless communication method should be developed to ensure communication between multiple sensors and the monitoring unit.

Therefore, it is necessary to develop an IoT-based environmental monitoring system capable of continuous collection and transmission of environmental and soil data from multiple cinnamon cultivation environments. This IoT component is aimed at solving the aforementioned problems through the use of ESP32-based sensor nodes and ESP-NOW protocol for communication. Collected data will be used further in the development of the research system and will serve for plant health assessment, growth monitoring, harvesting prediction, and automatic cinnamon harvesting.

1.4 Research Gap

Currently available systems for smart agriculture and environmental monitoring using Internet of Things (IoT) technology have already proven capable of collecting various environmental factors including temperature, humidity, soil moisture, and light levels. However, there is a lack of special solutions for cultivating cinnamon in several environmental conditions that incorporate not only environmental monitoring but also other intelligent factors like disease diagnosis, growth prediction, multispectral imaging, and harvesting. Additionally, most current systems either rely on standard Wi-Fi-based communications or are concerned only with monitoring one place of

cultivation, limiting their versatility in terms of the possibility of operating in distributed environments. Moreover, environmental data are usually considered separately from plant health and harvesting data. Thus, the need arises in creating an IoT architecture for distributed real-time monitoring and collection of data for cinnamon cultivation that will allow for collecting and transmitting environmental data from multiple sensors to the data collector and use it for the purposes of intelligent plant monitoring and harvesting.

1.4.1 Limited Cinnamon-Specific IoT Monitoring

Instead of precisely addressing the environmental monitoring needs of cinnamon farming, the majority of current IoT agriculture solutions are made for generic crops. A monitoring system that takes into account the particular environmental factors pertinent to cinnamon growth is required.

1.4.2 Multi-Environment Monitoring Gap

A single cultivation environment is the focus of many current methods. Distributed IoT platforms that can simultaneously monitor several cinnamon cultivation sites and compare environmental differences between them are lacking.

1.4.3 Real-Time Data Communication Gap

AI-based plant analysis and environmental sensor data are frequently gathered independently. The integration of IoT environmental data with multispectral analysis, growth prediction, disease detection, and harvesting readiness prediction is lacking.

1.4.4 IoT and AI/ML Integration Gap

AI-based plant analysis and environmental sensor data are frequently gathered independently. The integration of IoT environmental data with multispectral analysis, growth prediction, disease detection, and harvesting readiness prediction is lacking.

1.4.5 IoT and Automated Harvesting Integration Gap

There is little research that directly links an automated cinnamon harvesting system to environmental monitoring. Additional environmental data can be obtained by integrating IoT data with robotic harvesting, enabling more data-driven and informed harvesting decisions.

1.5 Aim and Objectives

1.5.1 Research Aim

This research component's primary goal is to create an Internet of Things (IoT)-based multi-environment monitoring system for cinnamon farming that allows for wireless transmission, real-time data collecting, and centralized monitoring of critical soil and environmental parameters. In order to enable the analysis of cinnamon plant growth and health, the system seeks to offer dependable environmental data from various cultivation conditions using ESP32-based sensor nodes and ESP-NOW communication. Additionally, the gathered IoT data is meant to support the

AI/ML, multispectral sensing, and automated harvesting parts of the entire research system, making the process of growing and harvesting cinnamon more intelligent, data-driven, and effective.

1.5.2 Research Objectives

1. To create an Internet of Things-based environmental monitoring system that can continually track critical soil and environmental factors related to cinnamon farming.
2. To create distributed sensor nodes based on ESP32 that can gather real-time data on temperature, humidity, soil moisture, and light/UV parameters from various agricultural settings.
3. To establish ESP-NOW wireless connection for dependable and effective sensor data transmission between the central data collector and the dispersed sensor nodes.
4. To provide a central data gathering system that can receive, recognize, arrange, and interpret environmental data from several sensor nodes.
5. To make gathered sensor data accessible for visualization, analysis, and historical monitoring in order to facilitate real-time environmental condition monitoring.
6. To test sensor data acquisition, wireless connection, data transmission, and data gathering under various environmental circumstances in order to assess the correctness and dependability of the built IoT system.
7. To incorporate IoT-generated environmental data with the entire research system, including the automated cinnamon harvesting equipment, multispectral sensing, disease detection, plant growth monitoring, and harvesting readiness prediction.
8. To offer a scalable Internet of Things framework that can be expanded with more sensor nodes or environmental factors to facilitate future advancements in automated harvesting and smart cinnamon cultivation.

1.6 Research Questions

RQ1: How can the ambient and soil conditions of cinnamon growing areas be continually monitored by an Internet of Things-based system?

RQ2: How well can temperature, humidity, soil moisture, and light/UV data from various cultivation conditions be gathered and processed by ESP32-based sensor nodes?

RQ3: How well can sensor data be sent from several dispersed sensor nodes to a central data collector using ESP-NOW communication?

RQ4: When gathering and sending real-time environmental data, what degree of accuracy and dependability can the designed IoT system achieve?

RQ5: How can the gathered environmental data be arranged, saved, and displayed to enable both historical and real-time monitoring of cinnamon cultivation conditions?

RQ6: How can plant health monitoring, disease detection, growth prediction, multispectral analysis, and harvesting readiness prediction be combined with environmental data supplied by the Internet of Things?

RQ7: How much can the limits of manual and sporadic environmental monitoring in cinnamon production be mitigated by the suggested IoT system?

RQ8: To what extent can the suggested IoT architecture be expanded to accommodate more sensor nodes, environmental factors, and future monitoring needs?

1.7 Scope, Limitations and Assumptions

1.7.1 Scope of Research

The design, development, deployment, and assessment of an Internet of Things-based environmental monitoring system for cinnamon farming are the main objectives of this study. The created method is meant to offer ongoing monitoring of critical soil and environmental factors that can affect the health and growth of cinnamon plants. Distributed sensor nodes are used to monitor various cultivation environments, while ESP32 microcontrollers serve as the main processing and communication devices in the Internet of Things component.

Temperature, humidity, soil moisture, light/UV conditions, and other important environmental data are all covered by the system. Every sensor node is in charge of gathering sensor readings, carrying out simple processing, and getting the information ready for wireless transmission. The use of ESP-NOW communication to move data from several sensor nodes to a central data collector is also covered in the study. Measurements from the dispersed nodes must be received, identified, arranged, and processed by the data collector.

The integration of the IoT component with the other parts of the entire cinnamon research system is also taken into account. AI/ML-based disease detection, growth monitoring, multispectral analysis, harvesting readiness prediction, and the automated cinnamon harvesting machine can all be supported by environmental data. However, rather than creating the AI models or the entire robotic harvesting system, the IoT component mainly concentrates on environmental data gathering, processing, transmission, and monitoring.

Lastly, testing and assessment of the created IoT system based on sensor data gathering, communication dependability, data transmission, data collection, and real-time monitoring performance are included in the scope. The goal of the project is to create a workable and scalable Internet of Things architecture that can be expanded in the future with more sensors, nodes, and monitoring features.

1.7.2 Limitations of the Research

- Only specific environmental data, such as temperature, humidity, soil moisture, and light/UV conditions, are monitored by the IoT system.
- The current implementation does not take into account other agricultural characteristics including soil pH, NPK levels, rainfall, wind speed, and specific soil composition.
- Environmental factors, hardware constraints, sensor placement, and calibration can all have an impact on the accuracy of sensor data.
- Physical barriers, distance, wireless interference, and the surrounding cultivation environment can all have an impact on ESP-NOW communication range.
- A small number of sensor nodes and cultivation are used in the system's development and testing.
- The results and specifications of the other study components determine how AI/ML disease detection, growth prediction, multispectral sensing, and robotic harvesting are integrated.
- Further study of network scalability, power management, sensor durability, communication dependability, and environmental protection would be necessary for large-scale deployment.

1.8 Individual Contribution

- created the Internet of Things architecture to monitor various surroundings used for cinnamon growing.
- ESP32 microcontrollers were chosen and incorporated as the primary processing and communication components.
- Temperature, humidity, soil moisture, and light/UV conditions are all measured by integrated environmental sensors.
- ESP32-based sensor nodes were created to gather and analyses environmental data.
- ESP-NOW communication was put into practice to transmit data wirelessly between dispersed sensor nodes and the central data collector.
- created a central data collector to receive, recognize, arrange, and handle data from several sensor nodes.
- Real-time environmental measurements were displayed and maintained through the implementation of data handling and monitoring systems.
- IoT data was integrated with the entire research system to provide environmental data that can assist the autonomous harvesting component, growth monitoring, multispectral analysis, illness detection, and harvesting readiness prediction.

1.9 Thesis Structure

Chapter 1 – Introduction

An overview of the planned study and each IoT-based environmental monitoring component is given in this chapter. The problem statement and identified research need are presented after the background and general context of the automated harvesting system and cinnamon cultivation. Additionally, it outlines the scope, constraints, assumptions, research questions, goals, and individual contributions of the study. An outline of the thesis's structure is given at the end of the chapter.

Chapter 2 – Literature Review

This chapter examines current IoT-based smart agriculture and environmental monitoring research, technologies, and systems. IoT applications in agriculture, environmental and soil monitoring, ESP32 microcontrollers, temperature and humidity sensors, soil moisture sensors, light/UV sensors, wireless sensor networks, and ESP-NOW communication are all covered. The benefits, drawbacks, and suitability of current systems for growing cinnamon are examined. The chapter also emphasizes the research deficit that the suggested IoT component aims to fill.

Chapter 3 – Research Methodology

The approach used to design, construct, and assess the Internet of Things-based environmental monitoring system is explained in this chapter. The research approach, requirements identification, hardware and software selection, sensor data collection techniques, ESP32 programming approach, wireless communication methodology, data collection procedure, and system evaluation methodologies are all explained. It also describes how the IoT component was created and incorporated into the larger research project's requirements.

Chapter 4 – System and Solution Design

The proposed IoT solution's detailed design is presented in this chapter. Distributed sensor nodes, ESP32 microcontrollers, environmental sensors, central data collectors, communication structures, and data flow are all covered. Additionally covered are the ESP-NOW communication network's architecture and the communication between sensor nodes and the data collector. To show how the suggested solution resolves the issues found, system workflows, component interactions, and design choices are described.

Chapter 5 – Implementation and Testing

The practical implementation of the suggested IoT system is described in this chapter. ESP32 sensor node development and programming, temperature, humidity, soil moisture, and light/UV sensor integration and configuration, ESP-NOW communication implementation, and central data collector development are all covered. Additionally, it displays how data processing and monitoring functionalities are configured. To confirm that the sensor nodes can gather

measurements and successfully send them to the central data collection, various functional and communication tests are carried out.

Chapter 6 – Results and Validation

The outcomes of the IoT system's implementation and testing are presented and assessed in this chapter. Analysis is done on sensor performance, wireless connectivity, data transmission, data collecting, and real-time monitoring. The constructed system's efficacy and dependability are assessed using the gathered data. Additionally, the chapter verifies if the deployed IoT component effectively accomplishes the specified study goals and offers valuable environmental data for the entire cinnamon farming system.

Chapter 7 – Discussion

The outcomes of the IoT system are examined in this chapter, along with their interpretation in light of the goals of the study and current methodologies. The performance of ESP-NOW communication, the efficacy of ESP32-based sensor nodes, the calibre of environmental data gathered, and the advantages of multi-environment monitoring are all covered. The difficulties encountered during development, system constraints, and the contribution of IoT data to other components like growth prediction, automated harvesting, AI/ML analysis, and plant health monitoring are also covered in this chapter.

Chapter 8 – Conclusion and Future Directions

The design, development, installation, and assessment of the IoT-based environmental monitoring system are summarized in this chapter, which wraps up the individual research contribution. It outlines the key accomplishments and describes how the created system supports ongoing, data-driven monitoring of cinnamon farming. The chapter also outlines potential future enhancements, such as expanding the system for larger-scale cinnamon cultivation environments, adding more environmental and soil sensors, increasing the number of sensor nodes, improving communication range and reliability, putting advanced power-management techniques into practice, and improving data visualization.

2. LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

The purpose of the literature study was to find and assess previous studies on wireless sensor networks, smart agriculture, IoT-based environmental monitoring, and real-time agricultural data collecting. The evaluation primarily concentrated on tools and techniques that can be utilized in the production of cinnamon, especially for soil and environmental monitoring. ESP-NOW, wireless connectivity, temperature and humidity sensors, light/UV sensors, soil moisture sensors, ESP32 microcontrollers, and centralized data collecting were all taken into consideration. Furthermore, relevant research on IoT-based crop monitoring and multi-environment agricultural monitoring was examined to comprehend current methods and their efficacy.

Research articles, conference papers, journals, technical publications, and other trustworthy scholarly sources around IoT and smart agriculture were taken into consideration for the review. The system design, sensors employed, communication technologies, data collection techniques, monitoring capabilities, and limits of the chosen studies were examined. Systems that allow numerous sensor nodes and offer real-time monitoring received special attention because these features are directly related to the suggested IoT component.

Limitations in current agricultural IoT systems, such as reliance on manual monitoring, restricted environmental factors, single-location monitoring, connectivity constraints, and lack of interaction with intelligent agricultural systems, were also included in the review's scope. For the suggested IoT-based multi-environment cinnamon monitoring system, the results of the examined literature were utilized to pinpoint the research gap and direct the choice of appropriate hardware, sensors, communication protocols, and system architecture.

2.2 Domain and Theoretical Foundations

The Internet of Things (IoT) and Smart Agriculture domains, where linked sensors and computer equipment are utilized to continually monitor agricultural settings and gather data for well-informed decision-making, are the foundation of the suggested IoT component. Environmental elements that affect plant growth and health in cinnamon production include temperature, humidity, soil moisture, and light levels. An appropriate method for continuously measuring these characteristics and sending the data gathered to a centralized monitoring system is provided by IoT technology. Sensor-based monitoring, wireless sensor networks, embedded systems, and real-time data communication form the theoretical basis of the suggested system.

ESP32 microcontrollers are used in the system as embedded communication and processing components. After receiving measurements from linked sensors and carrying out simple data processing, the ESP32 transmits the gathered data to the central data collector. It is possible to monitor several cultivation settings at once thanks to the distributed sensor-node technology. This adheres to the concepts of wireless sensor networks, in which a number of sensor nodes gather data from various physical places and send it to a gateway or central device.

The system uses ESP-NOW, a lightweight wireless communication protocol compatible with ESP32 devices, for wireless communication. Direct communication between ESP32 devices is made possible via ESP-NOW, which eliminates the need for a traditional Wi-Fi network connection. Because of this, it can send sensor measurements with comparatively little communication overhead between dispersed sensor nodes and the central data collector.

Real-time environmental monitoring and data collection, in which sensor measurements are gathered at predetermined intervals and transmitted for monitoring and analysis, are also included in the theoretical framework. The gathered data can be used in conjunction with other research components, such as AI/ML-based disease identification, growth prediction, multispectral analysis, and harvesting readiness prediction, to provide historical and current knowledge about the cultivation environment. As a result, the Internet of Things component creates the link between the system's digital analysis and decision-making processes and the actual cinnamon growing environment.

2.3 Related Research

The examined literature shows that various agricultural fields have looked into IoT-based agricultural monitoring, environmental sensing, wireless sensor networks, and real-time data collection. However, rather than being especially made for multi-environment cinnamon farming, these technologies are frequently created for general crop monitoring or specific agricultural factors. In order to determine current methods, technologies, and constraints pertinent to the suggested IoT component, the following papers were examined.

An IoT-based agricultural monitoring system for gathering environmental data including temperature, humidity, and soil moisture was studied by [Author et al. [1]]. The study shows how IoT technology may be used to continuously monitor agricultural surroundings and lessen reliance on manual observations. However, the system does not particularly address the environmental needs of cinnamon agriculture; instead, it mostly concentrates on general agricultural monitoring.

[Author et al. [2]] suggested an Internet of Things-based smart farming system that uses microcontrollers and sensors to gather environmental data in real time. The study shows how agricultural data for monitoring and decision-making can be continuously provided by sensor-based systems. Nevertheless, the suggested method does not offer a distributed monitoring architecture for several settings and concentrates on a restricted cultivation environment.

The application of ESP32-based sensor nodes for agricultural environmental monitoring was examined by [Author et al. [3]]. The study emphasizes how well ESP32 microcontrollers integrate various sensors, process measurements, and transmit gathered data. Since the ESP32 is also utilized as the primary processing and communication device, this research is pertinent to the suggested system. Nevertheless, the integration of ESP32 sensor nodes with a multi-environment environmental monitoring system is not the main focus of the study.

Wireless sensor networks for smart agriculture were studied by [Author et al. [4]]. The study shows how several dispersed sensor nodes may gather data from various locations and send the readings to a central monitoring unit. Since several sensor nodes are needed to monitor various cinnamon cultivation settings, this strategy is especially pertinent to the suggested IoT architecture. However, the deployment environment and node distance can affect communication scalability and dependability.

[Author et al. [5]] investigated the use of IoT-based sensors for soil moisture monitoring. The study shows how crucial soil moisture data is for comprehending agricultural growing conditions and assisting with decisions about irrigation. But rather of being coupled with other environmental measures like temperature, humidity, and light conditions, soil moisture is regarded as a separate parameter.

[Author et al. [6]] looked into wireless sensor technology for real-time environmental monitoring. The study shows that regular environmental data collection can yield more precise information than sporadic hand measurements. Nevertheless, neither harvesting-related decision-making nor the integration of environmental data with AI-based plant health evaluation are taken into account by the system.

[Author et al. [7]] looked into communication between ESP32 devices via ESP-NOW. According to the study, ESP-NOW can facilitate direct wireless communication between ESP32 devices without the need for a traditional Wi-Fi network connection between the nodes. This is essential to the suggested system since the central data collector receives sensor measurements from dispersed sensor nodes via ESP-NOW. However, more research is needed to ascertain its scalability and dependability in other agricultural settings.

A multi-node IoT monitoring architecture for agricultural settings was proposed by [Author et al. [8]]. The study shows how data can be gathered from multiple sensor nodes and sent to a central monitoring platform. Nevertheless, neither cinnamon cultivation nor the integration of environmental monitoring with plant disease diagnosis, growth prediction, multispectral analysis, and automated harvesting are particularly covered in the study.

Overall, the reviewed studies show that various agricultural and IoT applications have already looked into the individual technologies needed for the suggested IoT component, such as ESP32-based sensing, environmental monitoring, wireless sensor networks, ESP-NOW communication, and centralized data collection. Nonetheless, these methods are typically dispersed over many agricultural domains and systems. The development of a multi-environment IoT system specifically for cinnamon cultivation that continuously gathers environmental and soil parameters and uses this data as a supporting data source for disease detection, growth monitoring, multispectral analysis, harvesting readiness prediction, and automated cinnamon harvesting is a research gap. By combining dispersed ESP32 sensor nodes, several environmental sensors, ESP-NOW connectivity, and a central data collecting method into a single cinnamon cultivation monitoring system, the suggested IoT component fills this gap.

2.4 Existing Systems, Products and Techniques

IoT-based agricultural monitoring systems, environmental sensor monitoring systems, multi-node wireless sensor networks, ESP32-based monitoring systems, commercial smart farming products, wireless communication techniques, real-time environmental monitoring, and cloud-based agricultural data collection are some of the existing systems, products, and techniques that are pertinent to the suggested IoT component. These current methods show how IoT technologies can be used to gather soil and environmental data. They do not, however, particularly address multi-environment cinnamon cultivation or integration with disease detection, growth monitoring, multispectral sensing, and automated harvesting because many of them are made for general agricultural purposes.

2.4.1 IoT-Based Agricultural Monitoring Systems

Soil and environmental conditions in agricultural areas are frequently monitored using IoT-based agricultural monitoring devices. Microcontrollers and sensors are frequently used in these systems to gather environmental data, including temperature, humidity, soil moisture, light intensity, and others. After that, the gathered data is sent to a cloud platform or central monitoring system for analysis and visualization.

These devices show how IoT technology may offer continuous access to agricultural data and lessen reliance on manual environmental monitoring. Nevertheless, a lot of current systems are primarily focused on gathering environmental data and are made for broad crop monitoring. They do not directly integrate with disease detection, growth prediction, multispectral sensing, and automated harvesting, nor do they specifically take into account the environmental requirements of cinnamon farming. Thus, by creating an IoT monitoring system especially for multi-environment cinnamon cultivation, the suggested research expands on this idea.

2.4.2 Environmental Sensor Monitoring Systems

Various sensors are used by environmental sensor monitoring systems to continuously measure factors that impact plant growth. While soil moisture sensors are used to determine the amount of water present in the soil, temperature and humidity sensors are frequently employed to monitor atmospheric conditions. The amount of light that plants receive can also be tracked using light or UV sensors.

These systems offer helpful environmental data for comprehending shifts in crop circumstances. Systems that track just one or a small number of characteristics, however, might not be able to give enough information about the entire cultivation environment. Furthermore, it can be challenging to evaluate environmental variables across various production environments because sensor values are occasionally gathered from a single spot. By employing numerous sensor nodes to gather various environmental parameters from various locations, the suggested IoT component overcomes this constraint.

2.4.3 Multi-Node Wireless Sensor Systems

Agriculture frequently uses Wireless Sensor Networks (WSNs) to gather data from several dispersed sensing points. Every sensor node gathers environmental data and sends the values to a central gateway or receiver. This method eliminates the need for physical connections between individual sensor nodes, enabling the monitoring of large or dispersed crop areas. In addition to offering benefits for monitoring various environmental conditions, multi-node systems enable system expansion through the addition of more sensor nodes. However, when several nodes are installed, issues with data management, interference, network range, and communication dependability may arise. The suggested solution establishes communication between sensor nodes and the central data collector using a distributed ESP32-based architecture and ESP-NOW communication.

2.4.4 ESP32-Based Monitoring Systems

Because the ESP32 offers computing power, wireless connection, and compatibility for several sensor types in a comparatively small platform, ESP32-based monitoring systems have been extensively utilized in Internet of Things applications. ESP32 devices are capable of gathering sensor readings, processing them, and sending the information to a monitoring platform or another device.

Current ESP32-based agricultural systems show that the platform is appropriate for distributed sensing and inexpensive environmental monitoring. Nevertheless, a lot of implementations employ traditional Wi-Fi-based communication or concentrate on single-node monitoring. Instead of cinnamon production, they could also be designed for general agricultural settings. In order to enable multi-environment monitoring, the ESP32 is utilised as the primary controller of each sensor node in the proposed study. It is paired with several environmental sensors and ESP-NOW communication.

2.4.5 Commercial Smart Agriculture Products

For monitoring soil and ambient conditions, a number of commercial smart agricultural products are available. Temperature, humidity, soil moisture, weather monitoring, wireless gateways, and cloud-based dashboards are a few examples of these goods. These gadgets can give farmers direct access to environmental data and show how IoT technology can be used practically in agriculture.

2.4.6 Real-Time Environmental Monitoring Techniques

A key component of contemporary smart agriculture systems is real-time environmental monitoring. Sensors periodically or continuously gather environmental data and provide the readings to a monitoring system in place of physically taking measurements at predetermined intervals. This facilitates the faster identification of environmental changes and offers historical data.

Even though many current agricultural systems have real-time monitoring capabilities, the data

gathered is frequently mostly utilized for simple environmental observation. Real-time IoT data is a crucial input for the larger research system in the suggested system. Plant photos, multispectral data, AI/ML outputs, growth tracking, and harvesting readiness analysis may all be integrated with environmental parameters.

2.4.7 Centralized Data Collection and Monitoring

IoT systems frequently use centralized data collection to gather measurements from several sensor nodes and arrange the data for tracking and analysis. Data collected from various nodes can be identified by a central collector or gateway, which can then process the measurements and send them to a dashboard or other application components.

Because data from several places can be managed by a single system, this method is helpful for multi-environment monitoring. Nevertheless, current solutions might make use of internet-dependent architectures or cloud-based platforms, which aren't always appropriate for every research or field setting. Measurements sent from the ESP32 sensor nodes via ESP-NOW are received and arranged by a central data collector in the suggested system. This gives the suggested cinnamon monitoring system an organized layer for gathering data.

All things considered, the systems, goods, and methods that have been studied show that IoT may efficiently assist agricultural environmental monitoring. Nonetheless, the studied methods typically concentrate on specific technology or broad agricultural uses. Using distributed ESP32 sensor nodes, multiple environmental parameters, ESP-NOW communication, centralized data collection, and integration with AI/ML, multispectral sensing, and automated harvesting, a comprehensive IoT solution tailored for multi-environment cinnamon cultivation is still a crucial research area. By integrating these technologies into a single monitoring architecture for the larger cinnamon cultivation and automated harvesting system, the suggested IoT component fills this gap.

2.5 Gap synthesis and conceptual framework

2.5.1 Research Gap

A number of shortcomings in current IoT-based agricultural monitoring systems were found in the literature review. The majority of current systems are designed for general agricultural purposes rather than specifically for cinnamon production, even though many of them are capable of gathering environmental information including temperature, humidity, soil moisture, and light intensity. Additionally, current systems frequently concentrate on monitoring a single culture environment, which restricts their capacity to compare environmental conditions at several growing locations.

The administration and exchange of data from numerous dispersed sensor nodes is another highlighted issue. Although agricultural monitoring frequently uses wireless communication technologies like Wi-Fi, Bluetooth, Zigbee, and LoRa, little attention has been paid to a lightweight

ESP32-based multi-node design that uses ESP-NOW for direct communication with a central data collector. As a result, there is a chance to create a communication architecture that can effectively gather environmental data from several sensor nodes without relying on a traditional Wi-Fi network.

The integration of IoT environmental data with other intelligent agriculture system components is still lacking. Plant disease detection, growth monitoring, multispectral sensing, harvesting readiness prediction, and automated harvesting are not immediately linked to the data gathered by existing IoT systems, which frequently regard environmental monitoring as a separate function. Consequently, it's possible that the environmental data won't be fully utilized for wise decision-making.

Thus, the absence of an integrated, multi-environment IoT monitoring architecture created especially for cinnamon production is the primary research gap this study attempts to fill. ESP32-based distributed sensor nodes, several environmental sensors, ESP-NOW wireless communication, and a central data collector are all used in the suggested solution to close this gap. Plant health analysis, growth forecast, harvesting readiness, and automated harvesting can all be supported by the environmental data that has been gathered in conjunction with other research components.

2.5.2 Summary of the Research Gap

Table 1 Summary of Research Gap

Area	**Existing Research/System**	**Identified Limitation**	**Proposed Contribution**
Cinnamon-Specific Monitoring	Most IoT agricultural monitoring systems are developed for general crops.	Limited focus on the specific environmental monitoring requirements of cinnamon cultivation.	Develop an IoT monitoring system specifically designed to support cinnamon cultivation.
Multi-Environment Monitoring	Many systems monitor environmental conditions from a single location or cultivation area.	Difficult to compare environmental conditions across multiple cultivation environments.	Deploy multiple ESP32-based sensor nodes to monitor different cultivation environments.
Environmental Parameters	Existing systems commonly monitor selected parameters such as temperature,	Some systems provide limited environmental information by	Integrate temperature, humidity, soil moisture, and light/UV sensors

Area	Existing Research/System	Identified Limitation	Proposed Contribution
	humidity, or soil moisture.	monitoring only a small number of parameters.	within the IoT architecture.
Wireless Communication	Agricultural systems commonly use Wi-Fi, Bluetooth, Zigbee, or other wireless technologies.	Conventional communication may require network infrastructure and may not be suitable for the proposed distributed ESP32 architecture.	Implement ESP-NOW communication between distributed ESP32 sensor nodes and the central data collector.
Centralized Data Collection	Existing systems may use separate gateways, cloud platforms, or individual monitoring applications.	Data from multiple sensing locations may not be efficiently organized within a single research-specific system.	Develop a central data collector to receive, identify, organize, and process measurements from multiple sensor nodes.
Real-Time Monitoring	Many systems provide periodic or remote monitoring of agricultural parameters.	Limited integration of real-time environmental data with other intelligent system components.	Provide real-time environmental data collection and monitoring for the proposed cinnamon system.
IoT and AI/ML Integration	IoT monitoring is often implemented separately from AI/ML-based agricultural analysis.	Environmental data is not fully utilized with plant health, disease, and growth analysis.	Provide IoT data as an input to the wider AI/ML-based plant health and growth analysis components.
Multispectral Data Integration	Existing environmental monitoring systems generally focus on conventional environmental sensors.	Limited integration between environmental sensor data and multispectral plant information.	Enable environmental measurements to be considered together with multispectral information in the overall system.

Area	Existing Research/System	Identified Limitation	Proposed Contribution
Automated Harvesting Integration	Existing IoT systems mainly focus on monitoring and do not directly support automated harvesting.	Environmental monitoring is separated from harvesting readiness and robotic harvesting processes.	Provide environmental data to support harvesting readiness analysis and the automated harvesting component.
Scalability	Some systems are designed for a fixed number of sensors or a single monitoring location.	Expansion to additional cultivation environments or sensor nodes may require significant changes.	Design a distributed architecture that can be expanded with additional ESP32 sensor nodes and sensors.

3. RESEARCH METHODOLOGY

3.1 Research Design

In order to design, develop, implement, and assess an IoT-based environmental monitoring system for cinnamon cultivation, the research design for the IoT component takes a practical and experimental approach. The design's primary goal is to consistently gather crucial soil and environmental parameters from various production settings and produce trustworthy data for the larger research system. Distributed ESP32-based sensor nodes, wireless connectivity, environmental sensors, and a central data collector make up the Internet of Things system.

The first step in the study is to determine which environmental factors are crucial for keeping an eye on cinnamon farming. The primary characteristics are chosen to be temperature, humidity, soil moisture, and light/UV conditions. To create individual sensor nodes, appropriate sensors and ESP32 microcontrollers are integrated based on these specifications. Every node gathers sensor readings, completes preliminary processing, and gets the data ready for wireless transfer.

Several sensor nodes can function in various cultivation locations thanks to a distributed multi-environment design. The wireless communication protocol between the central data collector and the sensor nodes is called ESP-NOW. The data collector gathers measurements from the various nodes, determines the origin of each measurement, arranges the data, and makes it accessible for tracking and additional processing.

Functional and experimental testing are used to assess the developed system. To find reliability and performance problems, tests are conducted on sensor readings, wireless connectivity, data transmission, data gathering, and real-time monitoring. During testing, many communication scenarios and ambient factors are taken into account.The gathered data is utilized to assess whether the IoT component meets the stated goals of the study.

Lastly, the IoT data is seen as a component of the larger research system. Plant photos, multispectral data, AI/ML outputs, growth monitoring, harvesting readiness prediction, and automated harvesting may all be integrated with environmental parameters. As a result, the research design creates a link between the intelligent decision-making elements of the suggested cinnamon farming system and physical environmental monitoring.

3.2 Data, Participants or Study Materials

An environmental monitoring dataset gathered for cinnamon cultivation serves as the primary source of data for the IoT component of this study. The dataset includes daily measurements of significant soil and environmental factors that may have an impact on the growth and cultivation conditions of cinnamon plants. Temperature, humidity, and soil moisture are the primary factors taken into account in the dataset. These measures are utilized to give environmental data for the larger study system and to comprehend changes in the cultivation environment over time.

3.2.1 Cinnamon Environmental Monitoring Dataset

Table 2: Dataset Attributes

Data Category	Description
Dataset Name	Cinnamon Environmental Monitoring Dataset
Data Type	Environmental and soil monitoring data
Main Parameters	Temperature, Humidity, Soil Moisture
Data Frequency	Daily measurements
Data Start Date	01 January 2026
Data End Date	08 October 2026
Data Format	Date-based structured dataset
Main Purpose	Environmental monitoring and analysis for cinnamon cultivation
Application	IoT monitoring, plant growth analysis and wider system integration

3.2.2 Dataset Structure

Each record in the dataset represents the environmental conditions recorded for a certain day, and it is organized as a date-based environmental monitoring dataset. The date, temperature, humidity, and soil moisture are the primary characteristics. The temperature and humidity attributes reflect the atmospheric conditions of the cultivation environment, while the date property indicates the time the measurements were taken. The soil's moisture status is represented by the soil moisture property.

Individual environmental characteristics can be examined both individually and collectively thanks to the structured framework. Additionally, it enables the identification of changes in environmental conditions over time. As a result, the dataset can be utilized to find trends, variations, and connections between various environmental characteristics. Additionally, because of this structure, the data can be processed further and integrated with other parts of the suggested cinnamon monitoring system.

3.2.3 Data Period

The timeframe covered by the environmental monitoring data used in this study is January 1, 2026, to October 8, 2026. Throughout this time, daily measurements were kept in order to offer a continuous time-based depiction of the environmental conditions. Instead of depending on measurements from a single day or brief period of time, the research can detect changes and variations in temperature, humidity, and soil moisture over a longer period of time by using data.

The chosen time frame offers enough historical data to assist the assessment of environmental monitoring and pinpoint modifications to cultivation conditions. Additionally, the readings can be compared over time and utilized for additional research inside the planned IoT and broader smart agriculture system thanks to the date-based structure.

3.3 Experimental or Study Procedure

Data loading and early analysis, data pre-processing, feature selection and engineering, and environmental data analysis are some of the steps in the IoT component's experimental process. First, a structured dataset is created based on the dates of the temperature, humidity, and soil moisture readings. After that, the dataset is examined for inconsistent measurements, duplicate records, inaccurate entries, and missing values. The pertinent environmental features are chosen and prepared for additional analysis after the dataset has been prepared. After processing, the data can be utilized to find environmental trends and assist the larger system of study on cinnamon agriculture.

3.3.1 Data Loading and Initial Analysis

The environmental dataset was first arranged by recorded date after being put into the data analysis environment. Temperature, humidity, and soil moisture readings taken every day from the cinnamon growing environment are included in the dataset. To comprehend the composition and

caliber of the gathered data, a preliminary analysis was carried out. We looked at the quantity of records, data kinds, duplicate records, missing values, and the lowest and highest values of each environmental parameter.

Observing changes in the environmental parameters during the data collection period was another aspect of the initial investigation. To find variances and broad trends, daily temperature, humidity, and soil moisture data were compared. In addition to determining whether the gathered data was appropriate for additional processing and analysis, this stage assisted in identifying potentially aberrant results.

3.3.2 Data Pre-processing

Prior to additional analysis, data pre-processing was carried out to enhance the environmental dataset's consistency and quality. We looked for missing, duplicate, or inaccurately recorded values in the gathered records. If missing values were found, they were treated according to the kind and quantity of missing data. Additionally, the temperature, humidity, and soil moisture fields were examined to make sure that the values were recorded in the proper numerical formats and consistent units.

The dataset could be examined as a time-series dataset since the date field was standardized to preserve a constant chronological order. In order to separate real environmental differences from potential measurement errors, unusual or perhaps inaccurate sensor values were also examined. A cleaner and more dependable dataset was created as a result of these pre-processing procedures for further analysis and integration with the suggested IoT monitoring system.

3.3.3 Feature Selection and Engineering

The goals of the IoT component and the environmental factors pertinent to cinnamon growing were taken into consideration when choosing features. Temperature, humidity, and soil moisture were the primary features chosen from the dataset; the date was kept as the temporal feature. These characteristics were chosen because they offer crucial details regarding the cultivation environment's soil and ambient conditions.

The date and sensor measurements can be subjected to feature engineering to produce further data for study. For instance, if necessary, the date can be divided into day, month, or other time-based characteristics. The measurements gathered can also be used to determine daily variations, average values, minimum and maximum values, and correlations between temperature, humidity, and soil moisture. Within the overall research system, these constructed features can facilitate the investigation of environmental trends and offer valuable inputs for next analyses pertaining to plant development, health, and harvesting.

3.4 Measures and Evaluation Metrics

To assess the efficacy of environmental condition monitoring and categorization, the IoT component is measured quantitatively. Based on predetermined criteria for cinnamon cultivation, the gathered temperature, humidity, and soil moisture data can be categorized into environmental condition types, such as appropriate and unsuitable. The system's performance can then be assessed by comparing the projected condition with the expected or reference condition. The confusion matrix, precision, recall, accuracy, and false positive rate are the primary evaluation criteria taken into account. These metrics show how well the environmental data may be utilized to determine suitable and unsuitable production conditions.

3.4.1 Confusion Matrix

The categorization of environmental circumstances into groups like appropriate and inappropriate for growing cinnamon is assessed using a confusion matrix. It contrasts the environmental state as anticipated by the gathered sensor data with the actual state. True Positive (TP), True Negative (TN), False Positive (FP), and False Negative (FN) numbers make up the matrix. A correctly detected suitable condition is represented by a True Positive, and an appropriately identified unsuitable state is represented by a True Negative. A False Negative happens when a suitable condition is mistakenly identified as unsuitable, whereas a False Positive happens when an unsuitable state is mistakenly labelled as appropriate.

3.4.2 Precision

When the system anticipates a suitable condition, precision quantifies how well it recognizes such conditions. It shows the percentage of all conditions projected as appropriate that were correctly identified as suitable. The system generates fewer erroneous appropriate classifications when the precision value is higher.

Precision = TP / (TP + FP)

Precision for the suggested IoT system can show how accurately the environmental monitoring method detects conditions deemed appropriate for growing cinnamon.

3.4.3 Recall

Recall gauges the system's capacity to accurately recognized environmental circumstances that are truly conducive to the production of cinnamon. It shows the percentage of all actual suitable situations that were correctly identified. A higher recall means that fewer appropriate environmental situations are mistakenly categorized as inappropriate.

3.4.4 Accuracy

The aggregate percentage of environmental circumstances that are accurately identified is known as accuracy. It gives a broad picture of the classification performance and takes into account both correctly identified acceptable and unsuitable situations.

Accuracy = (TP + TN) / (TP + TN + FP + FN)

3.4.5 False Positive Rate

The percentage of inappropriate environmental circumstances that are mistakenly categorized as suitable is known as the False Positive Rate (FPR). The system is less likely to mistakenly classify inappropriate situations as suitable when the FPR is lower. This is helpful in assessing how reliable the classification of environmental conditions is.

$$\text{False Positive Rate} = FP / (FP + TN)$$

A low false positive rate is preferred in the suggested system since misclassifying an unsuitable cultivation condition as suitable could lead to an inaccurate environmental evaluation and potentially impact further studies within the entire cinnamon monitoring system.

3.5 Data analysis

To find trends, variances, and connections between the monitored parameters, the gathered environmental data is examined. The daily temperature, humidity, and soil moisture data gathered from the cinnamon growing environment are the main focus of the analysis. To comprehend the distribution and quality of the data, exploratory analysis is first carried out. The changes in environmental circumstances during the course of the data collection period are then identified using time-based analysis. In order to comprehend how changes in one parameter may be linked to changes in another, relationships between the environmental parameters are also investigated. The analysis's findings are helpful for assessing the IoT monitoring system and bolstering further studies on cinnamon farming.

3.5.1 Exploratory Data Analysis

To comprehend the general features of the gathered environmental dataset, exploratory data analysis, or EDA, is carried out. The amount of daily records, data kinds, missing values, duplicate records, minimum, maximum, and mean values, as well as changes in temperature, humidity, and soil moisture, are all examined in the study. The dataset's broad environmental conditions are identified using statistical summaries.

3.5.2 Environmental Parameter Distribution Analysis

The distribution of temperature, humidity, and soil moisture values over the course of the data collecting period is examined using environmental parameter distribution analysis. To determine the most prevalent environmental variables and fluctuations in the agricultural environment, the frequency and range of the recorded values are examined. Each parameter can be summarized using statistical measurements like mean, minimum, maximum, and standard deviation.

3.5.3 Environmental Monitoring Performance Analysis

The built IoT monitoring system's efficacy and dependability are assessed using environmental monitoring performance analysis. The analysis takes into account whether the temperature,

humidity, and soil moisture measurements are effectively collected by the sensor nodes and whether the data is accurately transmitted to the central data collector. Additionally, the received measurements' consistency and completeness are assessed.

3.5.4 Environmental Condition Distribution Analysis

Based on the gathered measures of temperature, humidity, and soil moisture, environmental condition distribution analysis is used to pinpoint times with various environmental conditions. The recorded values can be classified as appropriate, moderately suitable, or unsuitable conditions based on predetermined environmental parameters related to cinnamon growing.

To determine how often the cultivation environment stayed within the preferred ranges, the distribution of these conditions over the course of the data collection period can be analyzed. Understanding environmental changes and pinpointing times that might need attention can be accomplished with the use of this analysis. The information that is produced can also help with the broader research components that deal with plant health, growth tracking, and harvest readiness.

3.6 Ethics privacy security and responsible AI

The IoT component of this study collects, processes, stores, and analyses environmental data in accordance with ethical, privacy, security, and responsible data-handling principles. Temperature, humidity, and soil moisture data gathered for cinnamon farming make up the dataset used for the Internet of Things component. The dataset's privacy hazards are minimal because it doesn't include sensitive human-related data, identifying information, or personal information. Nonetheless, proper data management procedures are still adhered to in order to preserve the accuracy and consistency of the study data.

3.6.1 Data Privacy

Instead of personally identifiable information, the IoT collection includes soil and environmental parameters. The environmental conditions of the cinnamon cultivation environment are mostly represented by the recorded data. The IoT component's operation and analysis do not require personal information. As a result, no superfluous personal information is gathered or added to the dataset, which is treated as environmental study data. Only the intended research uses of the data—such as environmental analysis and aiding the larger cinnamon monitoring system—are carried out.

3.6.2 Data Security

When transmitting, gathering, and storing IoT data, security is taken into account. Using ESP-NOW communication, the ESP32 sensor nodes send environmental readings to the central data collector. The likelihood of unauthorized devices entering the sensor network is decreased by using appropriate device setup and restricted communication. Additionally, the gathered data must be kept in a secure setting and shielded from unwanted alteration or erasure.

3.6.3 Ethical Data Collection

The environmental measurements are meant to reflect the real circumstances of the cultivation environment and are gathered for research and system development. Installing and using sensors should not cause undue disruption to the cinnamon plants or their surroundings. Data should not be purposefully altered to yield beneficial research outcomes, and measurements should be documented consistently. During data processing, any missing, unusual, or perhaps erroneous sensor readings should be found and properly recorded.

3.6.4 Responsible AI and Data Usage

The broader AI/ML components of the suggested research system can employ the environmental dataset as an input. Therefore, while using IoT data for analysis linked to plant health, growth, or harvesting, responsible AI principles are taken into account. A cinnamon plant's health and readiness for harvesting can also depend on other elements, thus environmental measurements shouldn't be the main consideration when assessing its status. Thus, plant photos, multispectral data, and other pertinent information utilized by the system as a whole should be taken into account in addition to IoT measures.

3.7 Reproducibility and research artefacts

In order to facilitate the environmental monitoring method' repeatability, the Internet of Things component of this study was created and documented. Throughout the study, the hardware setup, sensor choice, ESP32 microcontroller configuration, communication technique, data gathering process, and data processing stages are all recorded. This documentation gives another researcher the knowledge they need to comprehend the development process and replicate an IoT monitoring setup.

The ESP32 sensor node programs, sensor configuration information, ESP-NOW communication implementation, central data collector software, and environmental monitoring dataset are the primary research artefacts. The dataset includes daily observations of soil moisture, temperature, and humidity taken between January 1, 2026, and October 8, 2026. To facilitate additional analysis, the dataset's structure—which includes the date and environmental measurement attributes—is constantly preserved.

The source code used to read sensor measurements, process the values gathered, send data across ESP32 devices, and receive and arrange measurements at the central data collector are examples of software artefacts. In order to replicate the communication and data gathering procedure under comparable circumstances, the setup and implementation details are provided. The development documentation also includes any updates, troubleshooting tasks, or significant configuration adjustments made during development.

Sensor readings, communication test findings, data collecting outcomes, and environmental analysis outputs are examples of testing and evaluation artefacts included in the research. These

artefacts enable a review of the experimental process and offer proof of the produced IoT system's performance. Screenshots, tables, graphs, and recorded observations are kept as supporting documentation for the research findings when appropriate.

Maintaining a uniform process for gathering and processing data further supports reproducibility. Future data collection can use the same sensor specifications, measurement structure, transmission strategy, and processing techniques. This preserves the fundamental design of the suggested system while enabling the IoT component to be expanded with more sensor nodes, environmental characteristics, or longer monitoring times.

4. PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The necessity to continuously monitor soil and environmental conditions in cinnamon growing areas led to the identification of the requirements for the proposed IoT component. The system uses dispersed ESP32-based sensor nodes to measure temperature, humidity, and soil moisture. ESP-NOW is used to send the collected data to a central data collector. To precisely specify what the system should do and the quality attributes that should be upheld, the requirements are separated into functional and non-functional requirements.

4.1.1 Functional Requirements

- Gathering Environmental Data - Measurements of soil moisture, temperature, and humidity will be gathered by the system from the linked sensors.
- Processing of Sensor Nodes - After receiving sensor readings, each ESP32 sensor node must carry out the necessary preliminary processing and get the measurements ready for transmission.
- Monitoring in Multiple Environments - In order to monitor environmental conditions from various growing environments, the system must support several sensor nodes.
- Data Transmission using Wireless - Using ESP-NOW, the system will send the gathered sensor measurements from the sensor nodes to the central data collector.
- Reception of Data - The distributed ESP32 sensor nodes will transmit measurements to the central data collector.
- Identification of Nodes - To discriminate between data from various contexts, the system must identify the source sensor node linked to each received measurement
- Organization of Data - The temperature, humidity, and soil moisture readings that are collected must be arranged in a systematic manner by the central data collector.

4.1.2 Non-Functional Requirements

- Reliability - Throughout the monitoring time, the system should consistently gather and send environmental measurements.
- Accuracy - Temperature, humidity, and soil moisture conditions should all be monitored by the sensors with reasonable accuracy.
- Communication Dependability - Reliable measurement transfer between sensor nodes and the central data collector should be possible with ESP-NOW connection
- Performance - For the system to enable near real-time environmental monitoring, sensor readings must be analyzed and communicated in a timely manner.
- Scalability - The architecture should enable the installation of more environmental sensors or sensor nodes without necessitating a total system redesign

4.2 Component Architecture within the Complete Solution

A number of interconnected parts make up the proposed AI-Driven Multi-Environment Cinnamon Growth Monitoring and Automated Harvesting System, which is intended to monitor cinnamon cultivation, assess plant conditions, forecast growth and harvesting readiness, and facilitate automated harvesting. The Internet of Things (IoT) component of this comprehensive solution offers the wireless connection layer and environmental data collection. It gathers soil and environmental measurements from various cultivation settings and makes the information accessible for additional analysis by other system components.

Distributed ESP32-based sensor nodes, environmental sensors, an ESP-NOW wireless communication system, and a central data collector make up the Internet of Things component. Depending on the applied sensor design, each sensor node measures specific environmental parameters such as temperature, humidity, soil moisture, and light or UV conditions. The measurements are processed and made ready for transmission by the ESP32. Without requiring each sensor node to connect via a traditional Wi-Fi network, ESP-NOW allows wireless communication between the sensor nodes and the central data collector.

After receiving measurements from the dispersed sensor nodes, the central data collector locates the relevant sources and arranges the data for monitoring and additional processing. The measurements gathered can be used to compare conditions in various cultivation environments and track changes in the environment over time. The investigation of changes in temperature, humidity, and soil moisture throughout the production of cinnamon is further supported by historical environmental data.

The other key elements of the entire research system are interconnected with the IoT component. Plant photos, multispectral sensor data, and the results of machine learning and artificial intelligence models can all be taken into account in addition to environmental measurements. Plant health and disease analysis, growth monitoring, and harvesting readiness prediction are all supported by this combined data. The automated cinnamon harvesting mechanism, which is

designed as a distinct part of the total system, can then benefit from the harvesting readiness information.

4.2.1 Overall Architecture

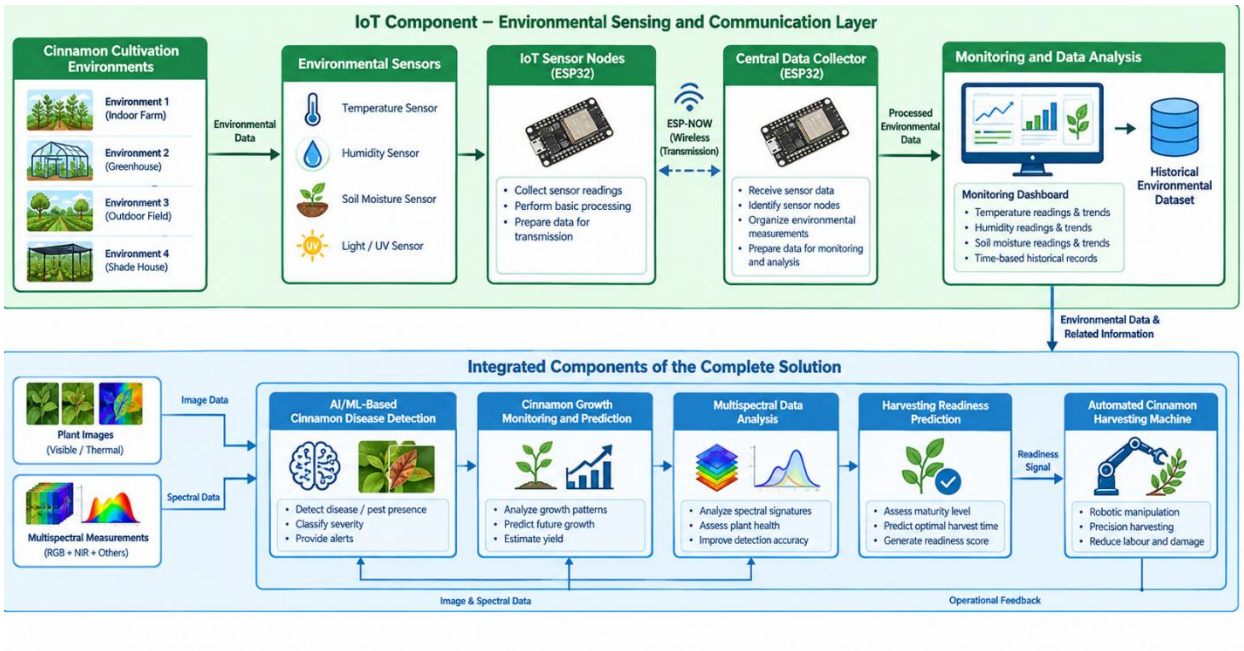


Figure 2: Overall Architecture

4.2.2 IoT-Based Environmental Sensing and Data Collection

The Internet of Things component gathers environmental data from various cinnamon growth settings, including shade houses, indoor farms, greenhouses, and open fields. Temperature, humidity, soil moisture, and light or UV levels are all measured by sensors in each setting.

Readings from various sensors are gathered by an ESP32 microcontroller, which also handles basic processing and gets the data ready for transmission. The gathered data is sent to a central ESP32 data collector via the ESP-NOW wireless communication protocol. The data collector locates the sensor nodes and arranges the environmental data for later use.

4.2.3 Environmental Monitoring and Data Analysis

A monitoring dashboard shows the environmental data that has been gathered. Users can track changes in environmental conditions over time by monitoring temperature, humidity, and soil moisture readings.

In order to facilitate future analysis, the system also saves historical environmental data. Researchers can analyses how various growing conditions impact cinnamon growth and comprehend environmental trends with the aid of these records. Additionally, the gathered data can support the system's AI-based components.

4.2.4 Integration with AI and Automated Harvesting Components

The IoT part collaborates with other parts of the entire cinnamon harvesting and monitoring system. Plant photos are used by the AI/ML-based disease detection component to find potential pest and disease issues. The growth monitoring component forecasts future growth by analysing growth trends.

In order to evaluate plant health, the multispectral data analysis component looks at observations taken from several spectral bands. To determine when cinnamon is suitable for harvesting, the harvesting readiness prediction component makes use of pertinent plant and environmental data. Lastly, based on the readiness data, the automated harvesting equipment harvests using robotic manipulation.

4.3 Design Alternatives and Decisions

In order to develop the suggested IoT-based environmental monitoring system, appropriate hardware, sensors, wireless communication protocols, network structures, and data gathering techniques have to be chosen. The needs for multi-environment cinnamon monitoring, dependable sensor data collecting, cheap installation costs, and integration with the whole AI-driven cinnamon growth monitoring and automated harvesting system were taken into consideration when making these choices. For every key design feature, a number of solutions were taken into consideration. The best choices were chosen based on their functionality, implementation difficulty, communication capabilities, and fit with the study's goals.

4.3.1 Microcontroller Selection

Sensor measurement collection, basic data processing, and environmental data transmission to the central data collector all depend on microcontrollers. Environmental monitoring systems can be developed using a variety of microcontrollers, such as ESP8266, ESP32, and Arduino boards. Arduino boards support a variety of sensors and offer a straightforward development environment; however, depending on the board chosen, extra modules may be needed for wireless connectivity. The ESP8266 can be used for simple Internet of Things applications and has built-in Wi-Fi connection. Nonetheless, the ESP32 has adequate processing power, several connectors for attaching sensors, and integrated wireless communication capabilities.\

4.3.2 Environmental Sensor Selection

The choice of environmental sensors is crucial since the study of cinnamon growth conditions is impacted by the precision and applicability of the readings gathered. Temperature, humidity, soil moisture, and light intensity can all be monitored using a variety of sensor types. The parameters to be measured, ESP32 compatibility, measurement range, precision, availability, and cost all influence the choice of sensors.

4.3.3 Wireless Communication Protocol Selection

To send measurements from dispersed sensor nodes to the central data collector, wireless connection is necessary. For this reason, a number of communication technologies can be taken into consideration, such as Wi-Fi, Bluetooth, Zigbee, and ESP-NOW. Although Wi-Fi facilitates network-based communication and is extensively utilized in IoT devices, it may necessitate additional equipment and network configuration. While Zigbee is intended for low-power wireless sensor networks, it may need additional configuration and compatible gear. Bluetooth is appropriate for short-range communication.

For connection between the central data collector and the ESP32-based sensor nodes, ESP-NOW was chosen. It makes it possible for compatible ESP32 devices to communicate wirelessly without the need for a traditional Wi-Fi network connection. This method facilitates direct connection between the sensor nodes and the receiver and lessens the requirement for extra network infrastructure. As a result, ESP-NOW is appropriate for the suggested distributed environmental monitoring architecture, even though radio interference, barriers, and distance still affect its communication range and dependability.

4.3.4 Sensor Network Architecture Selection

A sensor network design specifies the communication between several sensor nodes and the way measurements are gathered and sent to the central data collector. A mesh-based network, a distributed multi-node design, and a single-node architecture are examples of potential strategies. Although a single-node architecture is easy to set up, it could not offer enough coverage when keeping an eye on many cultivation settings. Although a mesh network can offer several communication channels, it adds complexity to network management and installation.

For the suggested system, a distributed sensor-node architecture was used. Every ESP32-based node gathers measurements from the sensors it is attached to, then uses ESP-NOW to send the information to the central data collector. This design offers a central place for receiving and organizing the measurements and facilitates the monitoring of several cultivation conditions. Subject to communication bandwidth and system setup, it also permits the incorporation of additional sensor nodes as needed.

4.3.5 Centralized Data Collection Approach

The system needs a way to receive, identify, and arrange the data after the sensor nodes gather measurements. Processing and displaying measurements separately at each sensor node is one option. Transmitting all measures to a central data collector, where they can be arranged for monitoring and additional analysis, is an additional choice.

The centralized strategy was chosen because it offers a single location for gathering data from several sensor nodes. In addition to organizing the environmental values and identifying the sources of incoming measurements, the central data collector can make the data accessible for monitoring and historical analysis. The future integration of environmental measures with plant

photos, multispectral data, and the results of other AI/ML components in the entire study system is also supported by this method.

4.3.6 Environmental Data Management and Monitoring

Environmental measures can be controlled by a combination of historical data gathering and real-time monitoring, preserved as historical records, or merely shown as current values. While displaying only current readings offers instant information, it makes it more difficult to examine long-term environmental changes. Trend analysis is supported by historical data storage.

The suggested system takes into account both historical data analysis and ongoing environmental monitoring. Temperature, humidity, and soil moisture measurements can be arranged with the dates or timestamps that correspond with them. Comparisons between various eras or agricultural conditions are made possible by this framework, which also facilitates the discovery of environmental variances. Additionally, trends can be examined and additional research on the circumstances of cinnamon cultivation can be supported by the historical record.

4.3.7 Summary of Design Decisions

Table 3: summary of design decisions

Design Area	**Alternatives Considered**	**Selected Design Decision**	**Justification**
Microcontroller	Arduino, ESP8266, ESP32	ESP32	Supports multiple sensor connections, data processing, and wireless communication.
Environmental Sensors	Single sensor, multiple sensors	Multiple environmental sensors	Enables monitoring of temperature, humidity, soil moisture, and light/UV conditions where implemented.
Wireless Communication	Wi-Fi, Bluetooth, Zigbee, ESP-NOW	ESP-NOW	Supports direct wireless data exchange between compatible ESP32 devices without requiring a conventional Wi-Fi network connection.
Network Architecture	Single-node, distributed nodes, mesh network	Distributed sensor-node architecture	Supports monitoring multiple cultivation environments.

Data Collection	Independent node-level processing, centralized collection	Central data collector	Provides a common point for receiving and organizing measurements from multiple nodes.
Data Management	Real-time readings only, historical records, combined approach	Real-time and historical data management	Supports current environmental observation and analysis of changes over time.
System Integration	Tightly coupled architecture, modular architecture	Modular integration	Separates IoT monitoring from AI/ML analysis and robotic harvesting functions.
Data Analysis	Individual parameter analysis, combined environmental analysis	Analysis of selected environmental parameters	Helps identify environmental variations relevant to cinnamon cultivation.
Scalability	Fixed single-node system, expandable multi-node system	Expandable distributed architecture	Allows additional sensor nodes to be considered for future development, subject to system capacity.
Maintainability	Centralized tightly coupled code, modular implementation	Modular implementation	Makes testing, troubleshooting, and future improvements easier

4.4 Detailed design

The hardware structure, sensor-node arrangement, wireless connection mechanism, central data gathering procedure, and environmental data management are all described in depth in the proposed IoT component's design. This design's primary goal is to create a dependable system for gathering environmental data from various cinnamon growing settings and sending it to a central place for monitoring and additional analysis. A central data collector, ESP-NOW wireless communication, environmental sensors, and ESP32 microcontrollers are all used in the design.

Each sensor node in the Internet of Things component gathers data from its networked sensors as part of a distributed monitoring system. After processing the readings, the ESP32 gets the data ready for wireless transmission. After receiving measurements from the sensor nodes, the central

data collector locates the sources of the data and arranges it for monitoring and past analysis. The primary design components of the suggested IoT component are explained in the ensuing subsections.

4.4.1 Hardware Design

The primary data collection device, power supplies, environmental sensors, and ESP32 microcontrollers make up the hardware design. Each sensor node's primary processing and communication component is the ESP32. To gather data pertinent to the conditions of cinnamon growth, Temperature, humidity, and soil moisture are among the chosen environmental factors; depending on the sensor setup used, light or UV readings may also be included. While soil moisture sensors gauge the soil's moisture content, temperature and humidity sensors provide details about the atmosphere. Additional data regarding the cultivation environment can be obtained from light or UV sensors.

4.4.2 Sensor Node Design

Every sensor node is built to carry out wireless connection, basic processing, and environmental data collection. The ESP32 receives the corresponding signals from the sensors, which gather measurements from the surrounding cultivation environment. Analogue or digital interfaces. After reading the sensor values and carrying out the required basic processing, the ESP32 arranges the measurements into an appropriate data format. The sensor node identifier, measured environmental variables, and, if available, a timestamp or measurement date may all be included in the prepared data. Incorporating a node identification facilitates the central data collector's ability to differentiate measurements obtained from various locations.

4.4.3 Wireless Communication Design

How sensor measurements are sent from the dispersed ESP32 nodes to the central data collector is specified by the communication design. The suggested system exchanges data between the sensor nodes and the receiver using ESP-NOW, a wireless communication protocol compatible with ESP32 devices.

The communicating ESP32 devices must be properly configured, including the necessary peer information and communication parameters, prior to data transmission. After preparing its measurement data, each sensor node transmits it to the receiver that has been configured. After receiving the sent data, the central data collector processes the incoming messages.

The goal of the communication design is to minimize the requirement for a traditional Wi-Fi network connection between the central data collector and the sensor nodes. However, a number of variables, including radio interference, physical barriers, transmission range, and device setup, affect dependable communication. To determine whether the chosen strategy satisfies the needs of the suggested system, communication testing is therefore necessary.

4.4.4 Central Data Collector Design

The IoT monitoring architecture's receiving and organizing unit is the central data collector. Its primary duty is to gather measurements that the dispersed sensor nodes transmit and prepare the data for monitoring and additional analysis.

After receiving a message, the collector extracts the environmental data and uses the available identification information to identify the originating node. In order to analyses and compare values from various nodes, the received data is then arranged into a consistent structure. Incomplete messages or erroneous measurements can also be found using appropriate validation.

The environmental monitoring system has a common point of collection thanks to the central data collector. Depending on the implemented system architecture, the arranged measurements can be shown, saved, or moved to a different program component. Additionally, this approach facilitates future integration with other parts of the automated harvesting and cinnamon monitoring system.

4.4.5 Environmental Data Structure Design

To handle environmental measurements obtained from various sensor nodes, a standardized data structure is needed. Fields like date or timestamp, sensor node identifier, temperature, humidity, and soil moisture are used by the suggested structure to arrange each measurement record. When the implemented hardware measures and transmits these parameters, light or UV values may be included.

A measurement or collection of measurements obtained from a specific sensor node is represented by each record. While the node identifier identifies the measurement source, the date or timestamp aids in determining when the measurement was taken. This method of data organization facilitates historical analysis, environmental comparisons, and the detection of changes in environmental conditions.

4.4.6 Data Processing and Monitoring Design

The data processing design describes the steps followed from sensor measurement to environmental monitoring. First, the sensors collect environmental measurements. Next, the ESP32 reads and processes the values before preparing the data for wireless transmission. The central data collector then receives the messages, identifies their sources, and organizes the measurements. The collected information can be used for real-time monitoring and historical analysis, depending on the implemented display and storage facilities. Current readings provide information about the observed cultivation conditions, while historical records help identify variations across different dates and environments. Basic data validation can help detect missing values, incomplete messages, and readings outside the expected measurement range.

4.5 Standards and professional practice

In order to facilitate dependable data collecting, maintainable software, safe hardware implementation, and responsible research, the construction of the suggested IoT-based environmental monitoring system adheres to pertinent technological standards and professional practices. Hardware compatibility, communication configuration, data integrity, software quality, and testing protocols must all be properly taken into account because the system makes use of ESP32 microcontrollers, environmental sensors, and wireless connectivity. These procedures make it possible to combine the IoT component with the other parts of the AI-driven cinnamon growth monitoring and automated harvesting system and guarantee that it functions consistently.

4.5.1 Hardware and Electrical Safety

Environmental sensors are connected to ESP32 microcontrollers using proper hardware design techniques. Before connecting components, it is necessary to verify sensor pin configurations, operational voltages, signal kinds, and power requirements. To prevent harming the microcontroller or related devices, special consideration should be paid to the ESP32's input voltage restrictions and the electrical needs of analogue and digital sensors.

4.5.2 Wireless Communication Standards and Practices

The central data collector and compatible ESP32 sensor nodes communicate wirelessly via ESP-NOW in the Internet of Things component. The ESP32 platform documentation's technical specifications and configuration recommendations should be adhered to during implementation. All communicating devices should have uniform definitions for device addresses, communication. To assess effective data delivery, message reception, and the impact of environmental or distance interference, communication testing should be carried out. The software should have appropriate handling for failed transmissions, missing messages, or communication disruptions as needed. The dependability of environmental data collecting is enhanced by these procedures.

4.5.3 Software Development and Coding Practices

A organized and maintainable programming methodology should be used to create the ESP32 firmware. When necessary, sensor-reading processes, data processing, message preparation, wireless transmission, and error handling should be arranged into distinct functions or modules. Reusable code, uniform formatting, meaningful variable names, and comments all enhance readability and make troubleshooting easier.

4.5.4 Data Accuracy and Integrity

To enable trustworthy analysis, environmental measurements must be gathered and handled consistently. Installing sensors in accordance with their functioning specifications and verifying the validity of their results are both important. Sensor measurements can be compared to reference readings to assess measurement accuracy when appropriate reference equipment are available.

The link between each measurement and the appropriate sensor node should be maintained by the central data collector. Information about dates or timestamps should be handled consistently, and entries that are erroneous or incomplete should be pointed out rather than being accepted as accurate measures. To facilitate comparisons, trend analysis, and future study, historical data should be kept in an organized manner.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The Internet of Things (IoT), artificial intelligence, machine learning, environmental data processing, and robotic manipulation technologies are all integrated into the proposed AI-Driven Multi-Environment Cinnamon Growth Monitoring and Automated Harvesting System. The Arduino IDE was used to program ESP32 microcontrollers in C++ to create the Internet of Things component. The ESP32 devices were equipped with environmental sensors to gather data on the circumstances of cinnamon growing. The central data collector and the dispersed sensor nodes were able to communicate wirelessly thanks to ESP-NOW.

For the pertinent IoT data integration requirements, Firebase was chosen as the database and cloud data management platform. When the implemented system is set up to communicate with the Firebase service, it can facilitate the store, synchronization, and retrieval of environmental measurements. Technologies for plant image analysis, environmental data analysis, growth monitoring, harvesting readiness prediction, and automated harvesting are also included in the broader research system. The primary programming languages, development environments, data processing technologies, machine learning libraries, and database technologies pertinent to the suggested solution are described in the next subsections.

5.1.1 Programming Language

The IoT component's firmware was created using C++. It was constructed on ESP32 microcontrollers to read measurements from environmental sensors, carry out simple data processing, and get data ready for wireless transmission. The programming skills needed to communicate with hardware interfaces, control sensor readings, and implement communication operations are provided by C++.

Depending on the linked sensor setup, the ESP32 firmware was designed to gather data on temperature, humidity, and soil moisture. Before being sent via ESP-NOW to the central data collector, the gathered values were arranged into an appropriate data format. Additionally, C++ facilitates the construction of error handling, communication callbacks, and other features necessary for the sensor nodes to work.

Python is an appropriate programming language for data analysis, image processing, and model construction for the broader AI and machine-learning components. Numerous libraries for machine

learning, data visualization, and numerical calculation are available with Python. The programming environment used for each AI/ML component determines how it is used throughout the entire system.

5.1.2 Development Environment

The ESP32-based Internet of Things component was primarily developed using the Arduino IDE. It offers a simple interface for C++ firmware creation, compilation, and uploading to microcontroller boards that are compatible. Additionally, the IDE facilitates serial monitoring, which enables developers to view sensor readings, examine communication messages, and diagnose hardware or software problems.

The Arduino IDE was used to setup the ESP32 board support package and the pertinent sensor and communication libraries in accordance with the hardware specifications. While ESP-NOW functionality was set up to transmit the gathered data between the sensor nodes and the central data collector, sensor-reading functionalities were created to obtain environmental measurements.

A Python development environment like Jupyter Notebook or Visual Studio Code can be used for dataset exploration, model construction, and evaluation for the broader machine-learning components. These tools offer an environment for testing machine learning models, displaying environmental trends, and executing data analysis algorithms. Only if they were truly utilized throughout the study may they be recognized as implemented development tools.

5.1.3 Data Processing Libraries

To convert unorganized sensor measurements into structured information that can be saved, viewed, and examined, data processing is necessary. The IoT component uses C++ functions and the appropriate ESP32 libraries to process data received by the central collector, organize transmitted messages, read sensor readings, and convert measurements into appropriate formats. Python tools like Pandas and NumPy are useful for organizing records, handling missing values, calculating descriptive statistics, and analyzing changes in temperature, humidity, and soil moisture over time in environmental dataset analysis. Graphs displaying daily environmental fluctuations and correlations between observed factors can be created using Matplotlib.

These Python libraries are suitable for research visualization and offline dataset analysis. They should only be included in the final implementation description if they were utilized in the actual data-processing workflow, as they are distinct from the embedded C++ firmware.

5.1.4 Learning Libraries

The entire research system can analyses cinnamon plant conditions, detect potential diseases, track growth, and forecast harvest readiness using machine-learning technology. Depending on the kind of data and the prediction objective, different libraries may be chosen. Deep learning models for plant image-based disease detection can be developed and assessed using TensorFlow. Image loading, resizing, preprocessing, and other image-related tasks are supported

by image-processing programs like OpenCV. Scikit-learn offers algorithms and assessment techniques for classification, regression, and the creation of exploratory models for structured environmental data.

Additional data for pertinent models can be obtained from environmental measurements gathered by the IoT component. For instance, when the system's data flow allows for such integration, records of temperature, humidity, and soil moisture can be examined in addition to plant photos and multispectral readings. Appropriate models chosen in accordance with the available dataset and research requirements can be used for growth prediction and harvesting. The AI/ML components of the larger research system can benefit from these libraries. The final thesis should only list the libraries and models that were really utilised; it should not suggest that a certain model was employed unless the project code and experimental findings support this.

5.1.5 Database

For the pertinent data management needs of the suggested system, Firebase was chosen as the cloud database platform. Depending on the database product and integration technique, a Firebase service may offer a central location for storing and retrieving environmental metrics. The measurement date or timestamp, sensor node identifier, temperature, humidity, and soil moisture values are examples of structured records.

Environmental measurements are gathered by ESP32 sensor nodes in the suggested IoT workflow and sent over ESP-NOW to the central data collector. The gathered measurements can then be uploaded for cloud storage and accessed by authorized system components if Firebase is coupled with the central collector or another network-connected application. Direct Firebase connectivity is not possible with ESP-NOW communication alone; data must be transferred to the cloud via a suitable internet-connected device, gateway, or application.

Firebase-based storage can facilitate integration with dashboards or other apps, historical monitoring, and the retrieval of prior measurements. Additionally, the saved environmental data may be utilized in conjunction with the AI/ML components when the entire system permits it, as well as in additional investigation of cinnamon growing conditions.

5.2 Component implementation

5.2.1 IoT-Based Environmental Monitoring Component

The suggested system's Internet of Things component is in charge of gathering and sending the environmental data needed to track the growth of cinnamon plants. To detect variables like temperature, humidity, and soil moisture, the system connects ESP32 microcontrollers to environmental sensors. When the appropriate sensors are used, light and UV measurements can also be incorporated.

Environmental measurements are gathered by the ESP32 sensor nodes from various monitoring sites. The gathered measurements are wirelessly transmitted from the sensor nodes to a central data collector using ESP-NOW. After receiving the readings, the central data collector arranges them based on the pertinent sensor node and measurement data.

When the central collector is linked to an appropriate network gateway or application, the gathered environmental data can be saved in a database like Firebase. After then, the stored data can be utilized for additional processing, historical analysis, and real-time monitoring. This method offers an organized way to monitor environmental conditions in cinnamon growth areas and lessens the need for ongoing manual measurements.

The AI/ML component of the suggested system additionally receives environmental data via the IoT component. These measures can be used to support cinnamon plant health analysis, growth projection, and harvesting-readiness assessment in conjunction with plant photos and multispectral data. As a result, the IoT component serves as the smart agriculture system's communication and data collection layer.

5.2.2 AI/ML-Based Cinnamon Plant Analysis Component

The suggested system's AI/ML component is intended to evaluate gathered data and facilitate wise decision-making in the production of cinnamon. Depending on the particular model and its accessible dataset, it collaborates with the IoT component by using multispectral data, plant photos, Images of cinnamon plants can be analyzed by the plant health and disease detection algorithm to find outward signs of plant diseases or health issues. Plant conditions can be classified and pertinent information extracted using machine learning or deep learning algorithms and image processing techniques. The amount, quality, and labelling of the training and testing photos determine the model's dependability.

To find patterns and project future growth trends, the growth prediction model can make use of past environmental observations as well as other pertinent plant-growing data. When appropriate data is available, parameters like temperature, humidity, and soil moisture can be used as input characteristics. This technique can be used to assess the relationship between cinnamon plant development and environmental factors.

By offering data on plant health, growth, and possible harvesting readiness, these models' outputs can assist the automated cinnamon harvesting component. The robotic harvesting system should only receive the models' predictions via the specified integration mechanism, and each model must be trained, tested, and verified independently.

5.3 Integration with the other components

The suggested AI-driven multi-environment cinnamon growth monitoring and autonomous harvesting system integrates the IoT component with the other elements. By gathering measurements like temperature, humidity, and soil moisture using ESP32-based sensor nodes, it

serves as the environmental data acquisition and transmission layer. The ESP-NOW communication protocol is used to send the gathered data to the central data collector, where it is received and arranged for monitoring and additional analysis. The gathered data can also be kept in Firebase for future analysis and retrieval if an appropriate network-connected gateway or application is put in place. This procedure offers the environmental data needed to track the circumstances of cinnamon growing in various settings.

The AI/ML component, which is intended to evaluate pertinent data and produce predictions, can be integrated with the gathered environmental data. Plant photos and multispectral data can help identify plant health and disease, while historical sensor readings can be utilized with appropriate plant-growing observations to enable growth prediction. The harvesting-readiness model may determine whether cinnamon plants are ready for harvesting by utilizing pertinent data and suitable plant traits. A more comprehensive picture of plant conditions can be obtained by combining environmental measures with image-based and multispectral data. However, each model's design, training data, and implementation determine its inputs and outputs, and the accuracy of predictions needs to be assessed through suitable testing.

Through the system's specified communication and control method, the outputs produced by the AI/ML models can assist the automated cinnamon harvesting component. For instance, the robotic system employs its own control logic and pertinent sensors to carry out the harvesting process, and harvesting-readiness predictions could be used as an input to the robotic harvesting workflow. The robotic arm is not directly controlled by the IoT component until a certain control interface has been put in place, but it does give environmental data. All things considered, combining IoT, AI/ML, multispectral sensing, and robotic manipulation results in a modular system where each part carries out a distinct task while supporting automated harvesting and enhancing cinnamon growth monitoring.

5.4 Verification and software testing

5.4.1 Sensor Functionality Testing

To assess how well the environmental sensors attached to the ESP32 microcontrollers worked, sensor functionality testing was done. Temperature, humidity, and soil moisture were the primary factors taken into account. To ascertain whether the sensors could gather ambient data and supply values to the ESP32 firmware, the sensor readings were analyzed. Inconsistent measurements, unexpected readings, and missing data were thought to be possible signs of sensor or connection problems.

5.4.2 ESP32 Firmware Testing

The goal of ESP32 firmware testing was to confirm that the program instructions were executed correctly. Sensor initialization, data collection, reading processing, and setting up environmental measurements for wireless transmission were all tested. In order to detect possible programming

problems and verify that the sensor nodes could carry out their intended data collection tasks, the firmware was also examined.

5.4.3 ESP-NOW Communication Testing

The evaluation of wireless data transmission between the ESP32 sensor nodes and the central data collector was the main goal of the ESP-NOW communication testing. The procedure entailed determining the source of the received data and determining whether the collector could obtain environmental measurements from the transmitting nodes. When assessing communication reliability, missing communications, unsuccessful transmissions, and communication disruptions were taken into account.

5.4.4 Central Data Collector Testing

The ability of the central data collector to receive, recognize, and arrange measurements sent by the ESP32 sensor nodes was assessed. To make sure the pertinent environmental factors were accurately represented, the received values were reviewed. The ability to separate and prepare data from various sensor nodes for further monitoring and analysis was another aspect of this testing.

5.4.5 Database Integration Testing

Verifying the transfer, storage, and retrieval of environmental measures when Firebase integration was employed was the main goal of database integration testing. The testing took into account whether gathered readings could be accessible for historical analysis and kept in the appropriate data format. Missing records and data consistency were also taken into account. This section could instead outline the intended integration and related testing requirements if Firebase integration was not finished.

5.4.6 Integration Testing

Examining how the IoT component interacts with the other parts of the suggested automated harvesting and cinnamon monitoring system was the main goal of integration testing. The primary factors taken into account were the compatibility of the pertinent data formats and communication interfaces, as well as the availability of environmental measurements for the planned AI/ML research. The implemented system architecture took into account integration with robotic harvesting, growth forecast, harvesting-readiness prediction, and plant health monitoring.

5.5 Deployment and operation

5.5.1 IoT Component Deployment and Operation

The IoT component is intended to function as the automated harvesting and cinnamon growth monitoring system's environmental monitoring layer. Environmental sensors are attached to ESP32 microcontrollers in order to gather data from the chosen monitoring environments, including temperature, humidity, and soil moisture. The Arduino IDE's C++ programming allows the sensor nodes to be set up to send measured data to a central data collector via the ESP-NOW

communication protocol. The central data collector gathers and arranges sensor node readings for monitoring and additional analysis during operation. The gathered data can also be moved to Firebase for storage and historical retrieval in cases where a network-connected gateway or application is used.

To get meaningful environmental measurements, the sensor nodes should be placed correctly within the cinnamon cultivation environments during deployment. Prior to continuous operation, the sensors, ESP32 connections, power supply, wireless connectivity, and data gathering procedure should all be inspected. The system gathers readings at predetermined intervals during regular operation and sends them to the central data collector. To find sensor malfunctions, power outages, communication breakdowns, and missing data, periodic inspection is necessary. The gathered environmental data can then be utilized to keep an eye on the circumstances of production and supply pertinent inputs to the AI/ML component. To facilitate upkeep, troubleshooting, and future growth, the deployment configuration needs to be recorded.

5.5.2 AI/ML Model Deployment and Operation

In order to enable cinnamon plant monitoring and harvesting decisions, the AI/ML component is designed to process pertinent data and produce outputs. The inputs could include multispectral data, photos of cinnamon plants, and environmental measures gathered by the IoT system, depending on the models that are put into practice. While the growth prediction model can forecast growth trends using past environmental and plant-growth data, the plant health and disease detection model may identify possible plant health issues by analysing appropriate photos. To determine if a plant is ready for harvesting, the harvesting-readiness model can assess pertinent plant traits and other input variables. The data must be sent to each model in the preprocessing structure and format needed for its development.

5.5.3 Integrated System Operation

The system as a whole integrates data transfer, AI/ML analysis, automated harvesting support, and environmental data collection. The IoT sensor nodes gather environmental data while in operation and send it to the central data collector. If the defined database architecture is put into place, the gathered data can be saved, accessed, and delivered to the pertinent models via the proper data interface. The AI/ML models evaluate the necessary inputs and produce outputs for growth forecasting, harvesting-readiness evaluation, and plant health monitoring. The appropriate application or robotic harvesting component can subsequently employ these outputs in accordance with the system's specified control procedure.

The proper operation of the sensors, ESP32 firmware, ESP-NOW communication, data management procedure, and AI/ML models are all necessary for reliable operation. To find missing sensor readings, communication disruptions, invalid inputs, and model prediction mistakes, routine checks should be carried out. Harvesting activities shouldn't be initiated only by an unconfirmed model forecast; instead, the robotic harvesting component should function in

accordance with its own control and safety criteria. All things considered, the deployment and operation approach facilitates the integration of intelligent analysis and IoT monitoring into a single cinnamon cultivation system while enabling the autonomous maintenance and improvement of individual components.

6 RESULTS AND VALIDATION

6.1 Research Results

6.1.1 IoT Environmental Monitoring Results

The suggested system's Internet of Things component focuses on gathering environmental measurements needed for monitoring cinnamon crop. The system gathers data on temperature, humidity, and soil moisture using ESP32-based sensor nodes. The environmental conditions in the agriculture regions under observation are revealed by the measurements that were gathered. Daily environmental records from January 1, 2026, to October 8, 2026, are included in the research dataset. These documents provide as a foundation for analyzing how environmental circumstances have changed over time and spotting trends that might be important for tracking cinnamon growth.

The availability and consistency of the recorded measurements, the existence of missing values, and the change in environmental conditions during the study period should all be taken into consideration when evaluating the findings of IoT research. The distribution of temperature, humidity, and soil moisture can be shown using graphs and descriptive statistics. These findings can be used to assess if the gathered data is appropriate for environmental monitoring and additional AI/ML component analysis.

6.1.2 Environmental Data Analysis Results

To comprehend differences in the measurements gathered and look into connections between environmental characteristics, environmental data analysis is utilized. To find times of variance and possibly inappropriate growing circumstances, temperature, humidity, and soil moisture can be examined separately and combined. When such data is available, historical records can also be utilized to compare measurements across various dates or monitoring conditions.

The analysis's findings may give the AI/ML models helpful environmental data. However, trustworthy reference ranges are needed for cinnamon production in order to determine what circumstances are appropriate or inappropriate. Rather than relying solely on environmental readings, any findings regarding how environmental variables affect plant development should be backed up by pertinent agricultural data or real plant-growing observations.

6.1.3 AI/ML Model Results

The goal of the AI/ML component is to enable cinnamon plant monitoring and harvesting decisions by analyzing pertinent environmental, image-based, and multispectral data. By contrasting its

predictions with validated plant image labels, the plant health and disease detection model may be assessed. By contrasting expected growth values with actual data, the growth prediction model may be assessed. Similarly, trustworthy harvesting-readiness evaluations can be used to compare the harvesting-readiness model.

6.1.4 Combined IoT and AI/ML Research Findings

Combining IoT environmental measurements with AI/ML analysis provides a broader approach to cinnamon cultivation monitoring. The IoT component supplies environmental information, while the relevant models analyze their required inputs to identify plant health conditions, estimate growth trends, or predict harvesting readiness. Plant images and multispectral data can provide additional information that is not available from environmental sensors alone.

The combined approach is intended to reduce dependence on isolated measurements and support data-informed cultivation decisions. The extent to which it improves monitoring or prediction must be determined through the actual experimental results. Therefore, the research findings should distinguish between the functions demonstrated by the implemented system and the benefits that remain to be validated.

6.2 Product and Technical Results

6.2.1 IoT Hardware Implementation Results

Environmental sensors, ESP32 microcontrollers, and a central data collector make up the IoT hardware component. The sensor nodes are intended to gather measurements from the monitored environments and prepare the readings for wireless transmission. The technical evaluation should look at sensor operation, reading availability, hardware connection stability, and consistency of measurements received by the central collector.

Photographs of the constructed sensor nodes, the connected sensors, and the central data gathering setup can be used to display the hardware findings. The implemented hardware components, their functions, and the observed operating conditions can all be summed up in a table. Claims about sensor accuracy should be backed up by comparison with a suitable reference device or another approved validation technique.

6.2.2 Software and Communication Results

C++ is used in the Arduino IDE to create the ESP32 firmware. The firmware is in charge of reading sensor readings, preparing the data, and sending it over ESP-NOW to the central data collector. The technical assessment should ascertain whether the receiver can recognize and arrange measurements from the pertinent sensor nodes and whether the firmware carries out the necessary Functions.

The number of transmission attempts, successfully received messages, packet-loss rate, and average transmission or response time—all of which have been measured—can be used to assess

communication performance. For instance, the percentage of transmitted messages that were unsuccessfully received can be used to compute the packet-loss rate. Actual test records, not presumptive performance figures, should be used to present the results.

6.2.3 Environmental Dataset and Database Results

The environmental dataset offers an organized log of daily measurements of soil moisture, temperature, and humidity from January 1, 2026, to October 8, 2026. The availability of the measures needed for analysis, data completeness, consistency, and proper formatting should all be taken into account while evaluating it technically. Before utilizing the dataset for model building, data preprocessing could be required to address missing records, duplicate entries, and erroneous values.

If Firebase was incorporated into the system that was put into place, it might be included in the final product. If so, the findings should show whether or not environmental measurements were successfully saved and retrieved. Firebase communication requires an appropriate network-connected gateway or application since ESP-NOW does not natively provide internet connectivity. Firebase integration should be recorded as a proposed or partially developed function if this connection was not made.

6.2.4 AI/ML Model Development Results

The models that were actually created and assessed should be described in the AI/ML product outcomes. The image dataset, the projected classes, the chosen algorithm, and the classification performance are examples of pertinent technical details for the illness detection model. The input features, predicted growth variable, prediction error, and comparison of projected and actual values are possible outcomes for the growth prediction model. The harvesting-readiness model can be assessed by displaying its validation findings, output categories or scores, and input features.

Each model's performance ought to be displayed in its own table or figure. Limitations including inadequate training data, unbalanced classes, missing plant-growth observations, or inadequate representation of cultivation settings should also be identified by the model's outputs. It is important to distinguish between models that have finished testing and those that are still in the design or development stage.

6.3 User or Stakeholder Validation

6.3.1 Cinnamon Farmer and Stakeholder Requirements

Stakeholder validation helps determine whether the proposed system addresses practical challenges associated with cinnamon cultivation. The project includes field-visit and farmer-feedback activities intended to understand cultivation practices, monitoring challenges, and the

information that may be useful for farmers. These observations can be compared with the proposed functions of the IoT monitoring and AI/ML components.

The validation should examine whether continuous environmental monitoring, historical data access, plant health analysis, growth prediction, and harvesting-readiness information address the needs identified during the research. Actual farmer responses and documented observations should be used to support the findings. General assumptions about farmer satisfaction should not be presented as confirmed feedback.

6.3.2 Validation of the IoT Monitoring Component

By determining whether the environmental indicators gathered are pertinent to monitoring cinnamon cultivation and whether the data is clear and helpful, the IoT component can be verified against stakeholder criteria. The utility of temperature, humidity, and soil moisture readings, the usability of the monitoring interface, and the capacity to examine past measurements are examples of pertinent evaluation criteria.

The actual questionnaire answers or interview notes might be used to summaries the comments from farmers or other stakeholders who tried the monitoring system. A results table, for instance, can display the proportion of respondents who thought the measurements were helpful, the monitoring features they liked, and any restrictions they pointed out.

6.3.3 Validation of the AI/ML Models

It is important to assess whether the AI/ML component's results are pertinent to the targeted agricultural tasks. Stakeholders with appropriate understanding of cinnamon production could assist in determining whether the results of disease detection are comprehensible, whether growth forecasts are significant, and whether harvesting-readiness outputs correspond with realistic harvesting considerations.

Quantitative model testing is not replaced by stakeholder perspectives. Appropriate labelled test data and pertinent assessment measures must be used to evaluate disease detection and prediction performance. Feedback from experts or farmers can offer supplementary proof on the model outputs' limitations, practical utility, and interpretability.

6.3.4 Stakeholder Feedback and Improvement Opportunities

Potential system enhancements can be found using input gathered from field visits, surveys, interviews, and demonstrations. The placement of sensors, the environmental parameters that are monitored, the way previous measurements are presented, the lucidity of model outputs, or the system's overall usefulness are all possible areas for development. Stakeholder findings should only be shared when improvements are backed by real input.

6.4 Integration Results

6.4.1 IoT Sensor Node and Central Collector Integration

A crucial component of the suggested monitoring architecture is the integration of ESP32 sensor nodes with the central data collector. The sensor nodes gather environmental data and send it to the collector via ESP-NOW. Integration testing should determine whether the collector gets the messages, locates the pertinent sensor nodes, and arranges the measurements according to a standard format. Communication logs, sample sensor readings, images, or pictures of the functional hardware can all be used to illustrate the outcomes. When several sensor nodes are used, the assessment should additionally look at how well the data from various nodes can be differentiated. Alongside the observed results, any communication gaps or missing readings should be noted.

6.4.2 Integration of IoT Data with AI/ML Models

In order to integrate the IoT and AI/ML components, models that use these parameters as input features must have access to pertinent environmental measurements. The data must be prepared in accordance with the input specifications of the chosen model and transferred via an installed interface. To minimize processing errors, timestamps, parameter names, units, and data formats should all be consistent.

6.5 Summary of findings

6.5.1 Key IOT Findings

The suggested cinnamon monitoring system's environmental data collection and communication infrastructure is provided by the Internet of Things component. While ESP-NOW facilitates wireless communication with the central data collector, ESP32-based sensor nodes are designed to gather environmental factors including temperature, humidity, and soil moisture. A foundation for examining environmental changes over time is provided by the daily environmental dataset from January 1, 2026, to October 8, 2026.

The actual sensor testing, data completeness, communication performance, and data gathering outcomes should all be summed up in the final findings. These findings will assist in determining whether the deployed IoT component satisfies its environmental monitoring specifications and pinpointing any lingering technological constraints.

6.5.2 Key AI/ML Findings

The AI/ML component is designed to help plant health analysis, growth prediction, and harvesting-readiness assessment by utilizing appropriate environmental, image-based, and multispectral data. Each model's conclusions must to be grounded in its real development and assessment outcomes. To illustrate the advantages and disadvantages of each developed model, pertinent performance metrics, dataset restrictions, and prediction mistakes should be given.

6.5.3 Key Integration and Validation Findings

The connections made between the sensor nodes, central data collector, database if implemented, AI/ML models, and automated harvesting component should be summarized in the integration findings. The findings should highlight any unfinished integrations and describe which data flows and interfaces were successfully tested. Findings from stakeholder validation should also include a summary of real farmer or expert input and an explanation of how it connects to the needs of the system.

7 DISCUSSION

7.1 Interpretation of Findings

7.1.1 Interpretation of IoT Environmental Monitoring Results

The ability of the IoT component to gather, transmit, and arrange environmental measurements pertinent to cinnamon growth should be taken into consideration while interpreting its results. The ESP32-based sensor nodes are intended to track variables including soil moisture, humidity, and temperature. These measures lay the groundwork for data-driven agriculture monitoring and offer information about environmental changes over time. Trends, variances, and any connections between the observed parameters can be examined using the daily environmental dataset that was gathered between January 1, 2026, and October 8, 2026.

Sensor accuracy, data completeness, and measurement consistency all affect how helpful the data is. The accuracy of later analysis may be impacted by missing readings, unexpected values, or communication disruptions. As a result, the findings of sensor and communication testing should be understood in conjunction with the results of environmental monitoring. Although judgements regarding cinnamon growth or plant health require appropriate agricultural reference data and pertinent plant observations, the data can aid in characterizing the observed environments.

7.1.2 Interpretation of AI/ML Model Findings

The goal of the AI/ML component is to convert pertinent data into information that is helpful for growing cinnamon. While the growth prediction model analyses pertinent historical data to anticipate growth patterns, the disease detection algorithm concentrates on recognizing possible plant health issues from appropriate plant photos. Based on the input features utilized during model building, the harvesting-readiness model is designed to determine whether a plant could be acceptable for harvesting.

These models' real evaluation findings must be used to interpret their efficacy. While MAE and RMSE can show the prediction mistakes of numerical growth models, classification metrics like precision, recall, and F1-score can help highlight strengths and shortcomings in disease identification. When a model is put to different cultivation sites or plant circumstances that have never been seen before, it may still face challenges even with promising test findings. As a result,

model design, dataset quality, and the conditions reflected in the assessment data should all be taken into account when evaluating model performance.

7.2 Comparison with Literature Baselines and Existing Solutions

7.2.1 Comparison with Traditional Cinnamon Monitoring Practices

Manual observation and recurring environmental condition measurements are the mainstays of traditional agricultural monitoring. The frequency and consistency of observations may depend on the availability of farmers and monitoring resources, even if these methods might yield valuable information. By using wireless data transfer and ESP32-based sensor nodes, the proposed IoT component seeks to enhance the availability and organization of environmental data.

7.2.2 Comparison with Existing IoT-Based Agricultural Systems

Applying this architecture to the environmental monitoring requirements of the larger cinnamon growth monitoring and automated harvesting project is the specific contribution of the suggested solution. The environmental parameters monitored, the number of monitoring locations, the communication mechanism, the data storage strategy, and the integration with predictive analysis are some of the elements that should be used to assess its appropriateness with pertinent published systems. To demonstrate how the suggested system differs from current methods, actual published studies should be quoted.

7.2.3 Comparison of Wireless Communication Approaches

Communication technologies like Wi-Fi, Bluetooth, Zigbee, LoRa, and other wireless protocols are used by various agricultural monitoring systems. The range, power consumption, network needs, data speed, and deployment complexity of these technologies vary. In the suggested IoT architecture, ESP-NOW was used for communication between the central data collector and the ESP32 sensor nodes.

Because ESP-NOW allows direct wireless communication between compatible ESP devices without requiring a traditional Wi-Fi access point for the device-to-device link, this decision is pertinent to the design. However, the amount of transmitting nodes, interference, device setup, distance, and environmental impediments all affect how well it works in practice. Therefore, measured communication performance and the needs of the desired culture environment should be the basis for comparison with other technologies.

7.3 Research contribution

7.3.1 Contribution to IoT-Based Cinnamon Environmental Monitoring

The design and development of an environmental monitoring strategy specific to the larger cinnamon farming system is the research's IoT contribution. While ESP-NOW offers a communication mechanism for sending values to a central data collector, ESP32-based sensor nodes are utilized to gather pertinent environmental measures. The arrangement of environmental

data from the chosen monitoring environments is supported by this architecture. Additionally, the study creates an organized environmental dataset that can be used to track variations in soil moisture, humidity, and temperature across time. The implemented hardware, data gathering records, and communication testing outcomes should all be used to illustrate the practical contribution. Its potential significance is in offering structured environmental data for monitoring and additional study on cinnamon growing.

7.3.2 Contribution to AI/ML-Based Cinnamon Analysis

The application of appropriate data analysis and prediction algorithms to cinnamon plant monitoring is the AI/ML contribution. These tasks could include harvesting-readiness evaluation using appropriate plant attributes or multispectral data, growth forecast using pertinent historical data, and plant disease diagnosis from photos, depending on the models actually constructed. Instead of treating the sensor system and prediction models as completely separate solutions, the research attempts to link these analytical activities with environmental monitoring. The degree of integration attained, valid model validation, and the availability of suitable datasets all affect how strong this contribution is. Any assertions on enhanced prediction performance need to be backed up by experimental data.

7.4 Product Contribution

7.4.1 IoT Monitoring Prototype

The implemented environmental monitoring hardware and its related firmware, when finished, make up the IoT product contribution. To gather and arrange measurements, the prototype makes use of ESP32 microcontrollers, wireless connection, environmental sensors, and a central data collector. In the chosen cinnamon cultivation areas, it offers a useful basis for environmental condition monitoring.

7.4.2 AI/ML Analysis and Prediction Functions

Each model's test results, sample inputs and outputs, and pertinent performance indicators should be used to illustrate its usefulness. Model projections should not be regarded as definitive diagnoses or verified harvesting choices, but rather as information that supports judgements. Which models have been tested, which have been put into practice, and which are still being developed should all be mentioned in the product description.

7.5 Limitations and threats to validity

7.5.1 IoT Sensor Accuracy and Data Quality

The calibration and quality of the environmental sensors determine the IoT component's dependability. Placement, exposure to the environment, hardware connections, power fluctuations, and changes in sensor performance over time can all have an impact on sensor readings. Environmental analysis and any AI/ML model that uses these measurements as input may be

impacted by inaccurate or missing measurements. A restricted range of environmental characteristics and a particular time period are also covered by the available dataset. Temperature, humidity, and soil moisture are helpful indicators, but they don't account for all the variables affecting cinnamon development. Stronger findings regarding plant health and growth may require additional agricultural observations and reference measurements.

7.5.2 Wireless Communication and Deployment Limitations

The distance between devices, physical barriers, radio interference, and the cultivation environment's operational parameters can all have an impact on ESP-NOW communication performance. In addition to limiting the quantity or distribution of sensor nodes that can be successfully supported in a given deployment, these issues may result in delayed or missing readings.

7.5.3 AI/ML Dataset and Model Limitations

The amount, quality, and representativeness of the available training and testing data determine how well the AI/ML models perform. Disease identification may be impacted by a lack of image variety, and growth prediction may be limited by a lack of past plant growth records. Reliable reference labels or professional evaluations that accurately reflect the pertinent harvesting circumstances are also necessary for harvesting-readiness prediction.

7.5.4 Integration and System-Level Limitations

The compatibility and dependability of the connections between sensor nodes, the central collector, any database service, AI/ML models, and the robotic harvesting component are critical to the operation of the entire system. The entire operation may be disrupted by disparities in data formats, missing inputs, poor communication, or software bugs. The functions that are actually implemented and tested should be taken into consideration while evaluating integration restrictions. For instance, if cloud storage is needed, a proper network-connected gateway or application is required because ESP-NOW does not immediately give internet access to Firebase. Similarly, until a suitable control interface and safe operating logic have been put in place, AI/ML forecasts cannot operate a robotic machine.

7.6 Implications

7.6.1 Implications for Cinnamon Cultivation

The suggested IoT monitoring strategy may offer a more organized method of observing environmental factors pertinent to the production of cinnamon. Records of temperature, humidity, and soil moisture from the past may be useful for identifying changes in the environment and reviewing circumstances over time. This knowledge may help cultivation management become more informed when paired with trustworthy agricultural advice.

7.6.2 Implications for AI/ML-Based Decision Support

By seeing trends in plant photos, environmental measures, and other pertinent data, AI/ML models could aid in agricultural decision-making. While growth and harvesting-readiness forecasts may offer more information for planning, disease detection may assist identify plants that need more examination. When the models are evaluated against trustworthy reference results and trained on representative data, these capabilities are most helpful. Field observation and agricultural knowledge should be complemented by the models rather than automatically replaced. When evaluating outcomes, prediction uncertainty, model constraints, and potential classification errors should be taken into account, especially when a decision could impact crop health or harvesting operations.

7.6.3 Implications for Future Research and Development

- IoT-based environmental monitoring for soil moisture, temperature, and humidity in cinnamon farming settings.
- ESP-NOW enables real-time wireless data transmission between the central data collector and ESP32 sensor nodes.
- Analysis of historical environmental data to find trends and differences in the conditions of cinnamon cultivation.
- Cinnamon plant photos are used for AI/ML-based plant disease identification.
- Plant growth data from the past and pertinent environmental parameters are used to anticipate growth.
- The automated cinnamon harvesting method uses harvesting-readiness prediction to assist in decision-making.
- IoT, AI/ML, robotic harvesting, and multispectral sensing are integrated to support smart agriculture and upcoming system enhancements.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

The goal of this research project was to integrate IoT technology, machine learning, multispectral detection, and robotic manipulation to create an AI-driven, multi-environment plant growth monitoring and automated harvesting system. The project's primary goal was to investigate how smart technologies may enhance environmental monitoring, plant health assessment, growth forecast, and harvesting-readiness evaluation in order to improve cinnamon agriculture.

Temperature, humidity, and soil moisture were among the critical environmental characteristics that were monitored by the individual IoT component. Sensor measurements were collected using ESP32 microcontrollers, and data transfer between sensor nodes and the central data collector was taken into consideration using ESP-NOW wireless communication. The gathered environmental data facilitates ongoing observation and offers a basis for examining environmental factors pertinent to the production of cinnamon. Identification of changes in growing circumstances throughout time can also be aided by historical environmental records.

In order to enable cinnamon plant health and disease detection, growth forecast, and harvesting-readiness assessment, the entire research system also integrates AI/ML-based techniques. Multispectral data and image-based analysis can offer more details about plant conditions. One way to lessen reliance on manual observation and promote more reliable agricultural decision-making is to integrate these technologies with an automated harvesting mechanism.

The project shows the potential benefits of integrating robotic automation, IoT monitoring, data analysis, and AI/ML methods into a single smart agriculture system. However, elements like sensor accuracy, communication dependability, dataset quality, model performance, and successful component integration all affect how effective the entire system is. Therefore, verifying the system's dependability and appropriateness for actual cinnamon cultivation requires both quantitative assessment and practical field testing.

In summary, this study lays the groundwork for future advancements in intelligent cinnamon harvesting and farming technology. Expanding environmental data collecting, boosting model performance, strengthening system integration, and assessing the solution under various cultivation conditions should be the main goals of future developments. These advancements may contribute to the creation of a smart agriculture system for cinnamon cultivation that is more dependable and scalable.

8.2 Achievement of objectives

This research project's primary goals were to create an IoT-based environmental monitoring component and integrate it with AI/ML methods to facilitate autonomous harvesting and cinnamon growth monitoring. The accomplishment of these goals is covered below.

- To assess temperature, humidity, and soil moisture in cinnamon production conditions, an ESP32-based Internet of Things environmental monitoring system was developed.
- To facilitate data transfer between ESP32 sensor nodes and the central data collector, ESP-NOW wireless communication was implemented.
- Gathered and arranged historical environmental data for additional study, such as date, temperature, humidity, and soil moisture readings
- Supported cinnamon plant disease detection, growth forecasting, and harvesting-readiness evaluation by integrating AI/ML techniques into the entire research system.
- Developed a comprehensive smart agricultural strategy that combines robotic harvesting, multispectral sensing, AI/ML analysis, and Internet of Things environmental monitoring.
- Future enhancements were identified, such as improving sensor accuracy, growing environmental datasets, assessing communication dependability, and carrying out additional field testing to gauge the system's usefulness.

8.3 Individual research and product contributions

8.3.1 IoT-Based Environmental Monitoring

The development of an IoT-based environmental monitoring component for cinnamon production was the main emphasis of the individual research contribution. Temperature, humidity, and soil moisture are among the key environmental factors that the system was intended to assess. These measures facilitate ongoing monitoring of environmental factors pertinent to cinnamon plants and offer valuable information about the growing environment.

8.3.2 ESP32 Sensor Node Development

The environmental sensor nodes' primary processing components were ESP32 microcontrollers. Before sending sensor readings to the central data collector, the sensor nodes were built to gather environmental measurements. This method facilitates the collecting of environmental data in a distributed manner and offers a basis for adding more sensors to the system.

8.3.3 ESP-NOW Wireless Communication Implementation

The wireless communication method between ESP32 devices was ESP-NOW. For connection between the sensor nodes and the central receiver, it facilitates local device-to-device data transmission without the need for a conventional Wi-Fi router. Within the suggested IoT monitoring architecture, this communication technique facilitates the transfer of environmental measurements.

8.3.4 Environmental Data Collection and Management

For additional analysis, environmental data was gathered, including measurements of soil moisture, temperature, humidity, and date. The period covered by the available dataset is January 1, 2026, through October 8, 2026. These documents serve as a foundation for analysing changes

in the environment over time and can encourage more research into the connection between environmental factors and the production of cinnamon.

8.3.5 Integration of IoT Data with AI/ML Models

Complementary capabilities are added to the whole research system by the IoT and AI/ML components. While the AI/ML models analyses pertinent photos, environmental recordings, or multispectral data in accordance with their specific input requirements, the IoT component offers environmental measurements. If the necessary data interfaces and integration procedures are put in place, combining these sources might offer a more thorough insight of cinnamon plant conditions.

8.4 Future directions

8.4.1 Expansion of Environmental Sensor Coverage

The IoT monitoring system should be expanded in the future by adding more environmental sensors as needed. Factors including light intensity, UV exposure, and soil nutrient levels could be taken into account in addition to temperature, humidity, and soil moisture. A more thorough understanding of the factors influencing cinnamon farming might be achieved by gathering a greater variety of environmental measures.

8.4.2 Improvement of Sensor Accuracy and Calibration

Future work should focus on improving sensor accuracy through regular calibration and comparison with reliable reference instruments. Sensor readings should also be checked for missing values, abnormal measurements, and inconsistencies. These improvements could increase the reliability of the environmental data collected by ESP32 sensor nodes.

8.4.3 Enhancement of ESP-NOW Communication Reliability

Future enhancements such as packet-loss monitoring, automatic retransmission when necessary, connection-status monitoring, and recovery mechanisms for communication failures could help determine whether the system can reliably transfer sensor measurements from dispersed nodes to the central data collector. The dependability of ESP-NOW communication should be assessed under various deployment and environmental conditions.

8.4.4 Improvement of Environmental Data Collection and Storage

The environmental information that is currently accessible includes measurements of soil moisture, temperature, humidity, and date that were taken between January 1, 2026, and October 8, 2026. By gathering measurements over longer time periods and under various cinnamon cultivation settings, future research should expand this dataset. Integration with Firebase or another appropriate database, when put into practice, could facilitate organized storage, access to previous data, and visualization via a dashboard for monitoring.

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GLOSSARY

Term	Definition
AI (Artificial Intelligence)	Technology that enables computer systems to perform tasks that typically require human intelligence.
AI/ML Model	A computational model used to identify patterns, make predictions, or classify data.
Automated Harvesting	The use of automated machinery or robots to perform harvesting tasks.
Cinnamon Growth Monitoring	The process of observing and evaluating environmental conditions and plant growth in cinnamon cultivation.
Central Data Collector	A device or system that receives and organizes data transmitted by sensor nodes.
Dataset	A collection of data used for analysis, model training, validation, or testing.
ESP32	A microcontroller platform used for sensor interfacing, data processing, and wireless communication.
ESP-NOW	A connectionless wireless communication protocol developed by Espresso for communication between supported devices.
Environmental Monitoring	The measurement and observation of environmental parameters such as temperature, humidity, and soil moisture.
Firebase	A cloud-based application development platform that can provide database and data synchronization services.
Humidity	The amount of water vapour present in the air.
IoT (Internet of Things)	A network of physical devices equipped with sensors and communication capabilities to collect and exchange data.
Machine Learning (ML)	A branch of AI that enables systems to learn patterns from data and make predictions or decisions.
Multispectral Sensing	A sensing technique that captures information from selected wavelength bands of electromagnetic radiation.

Plant Disease Detection	The identification of potential plant diseases using observable symptoms, images, or other relevant data.
Robotic Manipulation	The use of robotic mechanisms to move, position, or handle objects to perform specific tasks.
Sensor	A device that detects or measures a physical or environmental parameter.
Sensor Calibration	The process of comparing and adjusting sensor measurements against a reference to improve accuracy.
Soil Moisture	The amount of water contained in soil.
Temperature	A measurement indicating how hot or cold the environment is.
Data Collector Node	An IoT device that receives sensor readings from other devices and prepares them for further processing.
Wireless Communication	The transfer of data between devices without a physical communication cable.
Growth Prediction	The estimation of future plant growth using relevant historical and current data.
Harvesting-Readiness Prediction	The estimation of whether a plant is suitable for harvesting based on relevant characteristics and model inputs.
Model Training	The process of using data to teach a machine-learning model to recognize patterns.
Model Validation	The process of evaluating a model during development to help select appropriate model settings.
Model Testing	The evaluation of a trained model using separate data to assess its performance on unseen examples.
Precision	The proportion of positive predictions that are correct.
Recall	The proportion of actual positive cases correctly identified by a model.
F1-Score	A performance metric that combines precision and recall using their harmonic mean.

Real-Time Monitoring	The observation and presentation of measurements as new data becomes available.
Smart Agriculture	The application of technologies such as IoT, AI, sensors, and automation to improve agricultural monitoring and management.

APPENDICES

Appendix A: Source Code for the ESP32 Sensor Node

The ESP32 source code used to gather environmental data, such as temperature, humidity, and soil moisture, is included in this appendix. Sensor initialization, data collection, and processing logic provided for the Internet of Things monitoring component are all included.

Appendix B: Source Code for ESP-NOW Wireless Communication

The source code for ESP-NOW communication between ESP32 sensor nodes and the central data collector is provided in this appendix. Device initialization, message transmission, data receipt, and, when applicable, communication-related operations are all included.

Appendix C: Source Code for the Central Data Collector

The code created for the central data collector is included in this appendix. It explains the receipt, identification, organization, and preparation of environmental readings for storage or additional processing.

Appendix D – Environmental Dataset

The environmental dataset utilized in the study, which includes measurements of date, temperature, humidity, and soil moisture taken between January 1, 2026, and October 8, 2026, is presented in this appendix. The thesis may contain a sample of the dataset, and if necessary, the whole dataset may be supplied as a separate supplementary file.

Appendix E - IoT Hardware and Circuit Diagrams

The circuit designs, sensor connections, ESP32 pin layouts, and hardware setup photos used to create the environmental monitoring component are included in this appendix.

Appendix F – AI/ML Model Implementation and Outputs

The circuit designs, sensor connections, ESP32 pin layouts, and hardware setup photos used to create the environmental monitoring component are included in this appendix.